



openFinance API Framework

Data Dictionary

Version 2.2.1

21 October 2024

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1 Introduction

1.1 From Core XS2A Interface to openFinance API

With [PSD2] the European Union has published a directive on payment services in the internal market. Among others [PSD2] contains regulations on services to be operated by so called Third Party Payment Service Providers (TPP) on behalf of a Payment Service User (PSU). These services are

- Payment Initiation Service (PIS) to be operated by a Payment Initiation Service Provider (PISP) TPP as defined by article 66 of [PSD2],
- Account Information Service (AIS) to be operated by an Account Information Service Provider (AISP) TPP as defined by article 67 of [PSD2], and
- Confirmation on the Availability of Funds Service (FCS) to be used by a Payment Instrument Issuing Service Provider (PIISP) TPP as defined by article 65 of [PSD2].

To implement these services (subject to PSU consent) a TPP needs to access the account of the PSU. The account is managed by another PSP called the Account Servicing Payment Service Provider (ASPSP). To support the TPP in accessing the accounts managed by an ASPSP, each ASPSP has to provide an "access to account interface" (XS2A interface). Such an interface has been defined in the Berlin Group NextGenPSD2 XS2A Framework.

This XS2A Framework is now broadened to extended services and developed into a Version 2 API family. This interface is addressed in the following as **openFinance API**. This openFinance API differs from the XS2A interface in several dimensions:

- The extended services might not rely anymore solely on PSD2.
- Other important regulatory frameworks which apply are e.g. GDPR.
- The openFinance API can address different types of **API Clients** as access clients, e.g. TPPs regulated by an NCA according to PSD2, or corporates not regulated by an NCA.
- The extended services might require contracts between the access client and the ASPSP.
- While the client identification at the openFinance API can still be based on eIDAS certificates, they do not need to be necessarily PSD2 compliant eIDAS certificates.
- The extended services might require e.g. the direct involvement of the access client's bank for KYC processes.

Note: The notions of API Client and ASPSP are used because of the technical standardisation perspective of the openFinance API. These terms are analogous to "asset broker" and "asset holder" resp. in the work of the ERPB on a SEPA API access scheme.

Note: In implementations, the API services of several ASPSPs might be provided on an aggregation platform. Such platforms will be addressed in the openFinance API Framework as "API provider".

The following account access methods are covered by this framework:

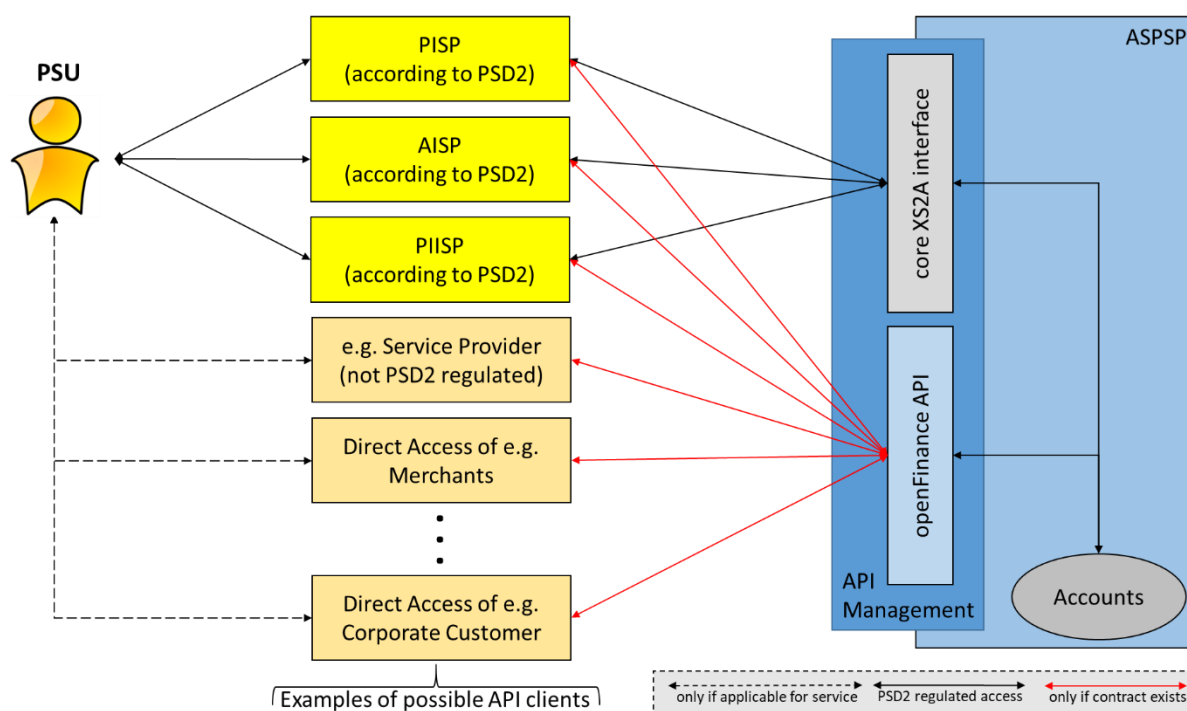


Figure 1: Core XS2A API and openFinance API

1.2 Data Dictionary for the openFinance API Framework

This document specifies the complex data structure to support the core XS2A API Services as well as the consent API specified for the openFinance API Framework. Also first V2 Premium Services are already covered.

Future V2 Premium Services will rely on this data dictionary as well, but might also define their own set of complex data structures which are not supported yet in the current version of the data dictionary.

1.3 Document Structure

This document covers complex data types in Section 2 and code lists in Section 3.

1.4 Document History

Version	Change/Note	Approved
2.0	Initial version of data dictionary for the openFinance API Framework Version 2	05 October 2023 openFinance TF
2.1	Updating the dictionary for attributes from Extended Payment Services, Extended Account Information Services, RTP Services, Push AIS Services, Admin Services as well as Discovery Services.	08 February 2024 openFinance TF
2.2	Updates related to openFinance API Framework Release 2.1 Details of the changes are covered in section 5.1	31 July 2024 openFinance TF
2.2.1	Introducing new data attributes and codes for the Mandate API. Updated Message Codes and some minor Errata Details of the changes are covered in Section 5.2	21 October 2024 oFA TF

Remark for Future: Please note that the Berlin Group openFinance API Framework is still under constant development. Technical issues, which are already in discussion within the Berlin Group openFinance working structure are mentioned in this document by "Remark for Future" to make the reader aware of upcoming potential changes.

2 Complex Data Types

In the following constructed data types are defined as used within the openFinance API Framework.

NOTE: Some of the data models have been changed from Version 1.3.x to Version 2.x of the openFinance API Framework. All attributes, where changes in subattributes applied, are marked by **yellow colour**.

NOTE: It is indicated below each data type, in which openFinance API Framework Implementation Guidelines the related data type is used in attributes of API requests and API responses on **first** level.

NOTE: The NISP project is a project platform to support NextGenPSD2 APIs and in future also openFinance APIs. This project platform had a.o. worked on an ASPSP testcase catalogue. This testcase catalogue is generic due to the potential variants of NextGenPSD2 APIs and so-called "Implementer Options" were defined for an ASPSP configuration spreadsheet to support derivation of test cases from the generic test case catalogue. During this derivation, the implementer options supported filtering the relevant test cases applicable to an implementation following the defined configuration. Since these Implementer Options are also used by ASPSP to partially document the PSD2 API implementations and since the NISP project has submitted some of the test case related work related to discovery services to the Berlin Group, the data types defined within the context of discovery services [oFA-Disc] below use "(IOx1, ... ,xn)" in the attribute description column to indicate that the addressed attribute is related to NISP Implementer Option 1 to NISP Implementer Option n, where $n \geq 1$. In future versions, these references will be deleted.

2.1 Account Access

Attribute	Type	Condition	Description
payments	Array of Account Access Rights	{Or Optional	Addressing required access rights to current accounts. The array may not be empty.
cards	Array of Account Access Rights	Or Optional	Addressing required access rights to single cards accounts. The array may not be empty.
cardAccounts	Array of Account Access Rights	Or Optional	Addressing required access rights to card reconciliation accounts. The array may not be empty.

Attribute	Type	Condition	Description
savings	Array of Account Access Rights	Or Optional	Addressing required access rights to saving accounts. The array may not be empty.
loans	Array of Account Access Rights	Or Optional	Addressing required access rights to loan accounts. The array may not be empty.
securities	Array of Account Access Rights	Or Optional}	Addressing required access rights to security accounts. The array may not be empty.

This data type is used in [oFA-Co].

2.2 Account Access Rights

The account access rights as posted by the API Client.

Attribute	Type	Condition	Description
account	Account Reference	Conditional	Is mandated if the consentType equals "detailed", not supported otherwise
rights	List of Access Rights Codes	Mandatory	

This data type is used in [oFA-Co] and by this document.

2.3 Account Access Rights Extended

The account access rights as provided by the ASPSP.

Attribute	Type	Condition	Description
account	Account Reference	Conditional	Provided after customer authentication if not already submitted by the API Client itself.

Attribute	Type	Condition	Description
rights	List of Access Rights Codes	Mandatory	
_links	Links	Optional	A hyperlink to the endpoint of the related account category.

This data type is used in [oFA-Co] and by this document.

2.4 Account Access and Authorisation Rights

Attribute	Type	Condition	Description
account	Account Reference	Mandatory	
rights	Array of Access Right Codes	Mandatory	Defines the reading rights on the related account identified above.
limits	Array of Limit	Optional	Defines the limits applicable only to this account.
paymentAuthorisations	Array of Product Rights	Mandatory	Defines authorisation related rights on level of dedicated payment products including specific limits. Empty array, if no payment right is provided.
paymentCancellations	Array of Product Rights	Mandatory	Defines payment cancellation rights on level of dedicated payment products. Empty array, if no payment cancellation right is provided.

This data type is used in [oFA-Disc].

2.5 Account Access and Authorisation Rights1

Attribute	Type	Condition	Description
account	Account Reference	Mandatory	

Attribute	Type	Condition	Description
rights	Array of Access Right Codes	Mandatory	
paymentAuthorisations	Array of Product Rights	Mandatory	Defines authorisation related rights on level of dedicated payment products including specific limits.

This data type is used in [oFA-Disc].

2.6 Account Access Extended

Attribute	Type	Condition	Description
payments	Array of Account Access Rights Extended	{Or Optional	Addressing required access rights to current accounts. The array may not be empty.
cards	Array of Account Access Rights Extended	Or Optional	Addressing required access rights to card accounts. The accounts can be card reconciliation accounts as well as single credit cards, dependent of the AIS offer of the ASPSP. The array may not be empty.
savings	Array of Account Access Rights Extended	Or Optional	Addressing required access rights to saving accounts. The array may not be empty.
loans	Array of Account Access Rights Extended	Or Optional	Addressing required access rights to loan accounts. The array may not be empty.
securities	Array of Account	Or	Addressing required access rights to security

Attribute	Type	Condition	Description
	Access Rights Extended	Optional}	accounts. The array may not be empty.

This data type is used in [oFA-Co].

2.7 Account Details

Remark: The ASPSP shall give at least one of the account reference identifiers listed as optional below.

Attribute	Type	Condition	Description
resourceId	Max70Text	Mandatory	This is the data element to be used in the path when retrieving data from a dedicated account.
iban	IBAN	Optional	IBAN of the account.
bban	BBAN	Optional	BBAN of the account. This data elements is used for payment accounts which have no IBAN.
msisdn	Max35Text	optional	An alias to access a payment account via a registered mobile phone number. This alias might be needed e.g. in the payment initiation services. . The support of this alias must be explicitly documented by the ASPSP for the corresponding API Calls.
other	Generic Account Identification	Conditional	An alias with a proprietary coding
currency	Currency Code	Mandatory	Account currency
ownerName	Max140Text	Optional	Name of the legal account owner. If there is more than one owner, then e.g. two names might be noted here. For a corporate account, the corporate name is used for this attribute.

Attribute	Type	Condition	Description
			Even if supported by the ASPSP, the provision of this field might depend on the fact whether an explicit consent to this specific additional account information has been given by the PSU.
ownerNames	Array of Account Owner	Optional	List of owner names together with a role, where applicable.
name	Max70Text	Optional	Name of the account, as assigned by the ASPSP, in agreement with the account owner in order to provide an additional means of identification of the account.
displayName	Max70Text	Optional	Name of the account as defined by the PSU within online channels.
product	Max35Text	Optional	Product Name of the Bank for this account, proprietary definition
cashAccountType	Cash Account Type	Optional	ExternalCashAccountType1Code from ISO 20022
status	Account Status Code	Optional	Account status. If this field is not used, than the account is available in the sense of this specification.
bic	BICFI	Optional	The BIC associated to the account.
linkedAccounts	Max70 Text	Optional	This data attribute is a field, where an ASPSP can name a cash account associated to the described account, e.g. in case of a securities account or a card account.
usage	Max4 Text	Optional	Specifies the usage of the account - PRIV: private personal account - ORGA: professional account

Attribute	Type	Condition	Description
details	Max500 Text	Optional	Specifications that might be provided by the ASPSP - characteristics of the account - characteristics of the relevant card
balances	Array of Balances	Conditional	
tariffs	Max1000Text	Optional	Currently only supported for securities accounts: Text information on tariffs related to this securities account.
applicableFees	Array of Securities Account Fee	Optional	Currently only supported for securities accounts: Array of elements, each representing one fee (rule) that is individually applicable for this Securities account. If, for example, an ASPSP applies a standard "managementFee" X, but the consumer negotiated a discount D, an element should represent (X-D), but neither X nor D.
interest	Array of Interest	Optional	Currently only supported for savings accounts and loan accounts: A set of interest rates associated to this account, e.g. interests for savings or credits.
relatedDates	Account related Dates	Optional	Currently only supported for savings accounts and loan accounts: Dates related to this account, e.g. start Date of a loan in case of a loan account.
collateralsInvolved	Boolean	Optional	Currently only supported for loan accounts: Is set to true if a collateral is involved securing the loan (e.g. security deposit, physical collaterals like houses, cars). More information might be provided under the "details" attribute.

Attribute	Type	Condition	Description
guaranteeInvolved	Boolean	Optional	Currently only supported for loan accounts: Is set to true in case of specific collaterals where a guarantee of payment of the loan is given by a different person.
_links	Links	Optional	Links to the account, which can be directly used for retrieving account information from this dedicated account. Links to "balances" and/or "transactions" These links are only supported, when the corresponding consent has been already granted.

This data type is used in [oFA-IG-Com], [oFA-IG-EAIS].

2.8 Account Owner

Attribute	Type	Condition	Description
name	Max70Text	Mandatory	Account owner name
role	Max35Text	Optional	The following proprietary codes are used: "owner", "legalRepresentative", "authorisedUser"

This data type is used in [oFA-IG-Com].

2.9 Account Reference

This type is containing any account identification which can be used on payload-level to address specific accounts. The ASPSP will document which account reference type it will support. Exactly one of the attributes defined as "conditional" shall be used.

Remark: The currency of the account is needed, where the currency is an account characteristic identifying certain sub-accounts under one external identifier like an

IBAN. These sub-accounts are separated accounts from a legal point of view and have separated balances, transactions etc.

Attribute	Type	Condition	Description
iban	IBAN	Conditional	
bban	BBAN	Conditional	This data elements is used for payment accounts which have no IBAN.
pan	Max35Text	Conditional	Primary Account Number (PAN) of a card, can be tokenised by the ASPSP due to PCI DSS requirements.
maskedPan	Max35Text	Conditional	Primary Account Number (PAN) of a card in a masked form.
msisdn	Max35Text	Conditional	An alias to access a payment account via a registered mobile phone number.
other	Generic Account Identification	Conditional	An alias with a proprietary coding
typeCode	Cash Account Type	{Or - Optional}	Remark: Shall not be used in the consent model, since account category provides enough information for the ASPSP.
typeProprietary	Max35Text	Or – Optional}	Remark: Shall not be used in the consent model, since account category provides enough information for the ASPSP.
currency	Currency Code	Optional	ISO 4217 Alpha 3 currency code
proxy	Proxy Account Identification	Conditional	

This data type is used in [oFA-IG-Com], [oFA-PFDom]. [oFA PAIS].

2.10 Account Reference1

Attribute	Type	Condition	Description
iban	IBAN	Conditional	
bban	BBAN	Conditional	This data element is used for payment accounts which have no IBAN.
pan	Max35Text	Conditional	Primary Account Number (PAN) of a card, can be tokenised by the ASPSP due to PCI DSS requirements.
maskedPan	Max35Text	Conditional	Primary Account Number (PAN) of a card in a masked form.
currency	Currency Code	Optional	ISO 4217 Alpha 3 currency code

This data type is used in [oFA-PFDom].

2.11 Account Reference2

Attribute	Type	Condition	Description
iban	IBAN	Mandatory	
currency	Currency Code	Optional	ISO 4217 Alpha 3 currency code

2.12 Account Reference3

Attribute	Type	Condition	Description
iban	IBAN	Conditional	
bban	BBAN	Conditional	This data element is used for payment accounts which have no IBAN.
other	Generic Account Identification	Conditional	An alias with a proprietary coding

Attribute	Type	Condition	Description
currency	Currency Code	optional	ISO 4217 Alpha 3 currency code

This data type is used in [oFA-PFDom].

2.13 Account Reference4

Attribute	Type	Condition	Description
iban	IBAN	Optional	
other	Generic Account Identification	Optional	

2.14 Account related Dates

Attribute	Type	Condition	Description
contractStartDate	ISODate	Optional	Start of the account contract.
contractEndDate	ISODate	Optional	End of the account contract.
contractAvailabilityDate	ISODate	Optional	Next Date where money can be taken off a saving account without loss of interest, if initiated now (in case of a saving account). Next Date where a loan can be fully repaid without any extra fee.

2.15 Account Report

Attribute	Type	Condition	Description
booked	Array of transactions	Conditional	Shall be contained if bookingStatus parameter is set to "booked", "both" or "all".

Attribute	Type	Condition	Description
pending	Array of transactions	Optional	Not contained if the bookingStatus parameter is set to "booked" or "information".
information	Array of transactions	Optional	Only contained if the bookingStatus is set to "information" or "all" and if supported by ASPSP.
_links	Links	Mandatory	The following links might be used within this context: • account (mandatory) • first (optional) • next (optional) • previous (optional) • last (optional)

This data type is used in [oFA-IG-Com].

2.16 Account Resource Reference

Attribute	Type	Condition	Description
resourceId	UUID	Conditional	To be used in authorisation processes, when updating a resource by an account.

This data type is used in [oFA-PFSM]

2.17 Accrued Interest

Represents the accrued interest by a securities position:

Attribute	Type	Condition	Description
daysAccrued	Number	Optional	Specifies the number of days used for calculating the accrued interest amount.
amounts	Array of Amount	Optional	Amount of the accrued interest. Each item represents

Attribute	Type	Condition	Description
			the same monetary value in different currencies, e.g. account currency, currency of the security's denomination.

2.18 Additional Party Information

Attribute	Type	Condition	Description
tradeName	Max70Text	Optional	Trade name of the related party. For display to the PSU only.
merchantCategoryCode	Merchant Category Code	Optional	Merchant Category Code as assigned by ISO:
logoUrl	Max2048 Test	Optional	A hyperlink to the logo of the party.
geoLocation	GEO Location	Optional	
merchant Logo	String	n.a.	Base 64 coded graphic. Reserved for future extensions.

2.19 Additional Request Information

Attribute	Type	Condition	Description
paymentConditions	Payment Conditions	optional	Conditions for the execution of the payment.
creationDateTime	ISODateTime	optional	creationDateTime of the originator
requestedExpiry DateTime	ISODateTime	mandatory	Date and Time at which this request expires.
originatorPspId	Financial Institution Identification	Optional	Originator payment service provider in case of indirect participation regarding

Attribute	Type	Condition	Description
			a related API Access Scheme or bilateral contract
receiverPspId	Financial Institution Identification	Conditional	Receiving payment service provider as defined by an API Access Scheme or bilateral contracts in case of not addressing the receiving institution directly. It is mandatory in case where the technical API Provider is not the receiver as such.

2.20 Additional Request Information1

Attribute	Type	Condition	Description
originatorPspId	Financial Institution Identification	Optional	Originator payment service provider in case of indirect participation regarding a related API Access Scheme or bilateral contract

This data type is used in [oFA-Co], [oFA-PFDom].

2.21 Address

Attribute	Type	Condition	Description
streetName	Max70Text	Optional	
buildingNumber	String	Optional	
townName	String	Optional	
postCode	String	Optional	
country	Country Code	Mandatory	

2.22 Agent Description

Attribute	Type	Condition	Description
financialInstitutionId	Financial Institution Identification	mandatory	
branchIdentificationId	Max35Text	Optional	Unique and unambiguous identification of a branch of a financial institution.
branchIdentificationName	Max140Text	Optional	Name by which an agent is known and which is usually used to identify that agent.

2.23 Agent Description1

Attribute	Type	Condition	Description
financialInstitutionId	Financial Institution Identification	mandatory	

This data type is used in [oFA-PFDom].

2.24 Agent Description2

Attribute	Type	Condition	Description
financialInstitutionId	Financial Institution Identification1	mandatory	

This data type is used in [oFA-PFDom].

2.25 Amendment Information Details

Attribute	Type	Condition	Description
originalMandateId	Max35Text	Optional	Mandatory of change occur in "Mandate Identification" otherwise not to be used.

Attribute	Type	Condition	Description
originalCreditorSchemeld	Party Description4	Optional	Mandatory of change occur in "Creditor Scheme Identification" otherwise not to be used.
originalDebtorAccount	Account Reference4	Optional	Mandatory of change occur in "Debtor Account" otherwise not to be used.
originalDebtorAgent	Agent Description2	Optional	If changes occur in debtor agent and SMNDA is not used in "originalDebtorAccount"

2.26 Amount

Attribute	Type	Condition	Description
currency	Currency Code	Mandatory	ISO 4217 Alpha 3 currency code
amount	String	Mandatory	The amount given with fractional digits, where fractions must be compliant to the currency definition. Up to 14 significant figures. Negative amounts are signed by minus. The decimal separator is a dot. Example: Valid representations for EUR with up to two decimals are: • 1056 • 5768.2 • -1.50 • 5877.78

This data type is used in [oFA-IG-Com], [oFA-PFSM], [oFA-IG-EPIS]., [oFA-IG-RTP].

2.27 Amount dependent Rate

Attribute	Type	Condition	Description
percentage	String	Mandatory	The (signed) rate of the interest. Up to 20 significant figures. The decimal separator is a dot. Trailing zeroes must be truncated.
fromAmount	Amount	Optional	Amount from which the interest rate applies.
toAmount	Amount	Optional	Amount up to which the interest rate applies.

Remark: ISO 20022 is defining Percentage as a decimal type, where 5.0 expresses 5% interest rate. This specification is using a signed percentage to be able to express also negative percentage codes, i.e. -1.2 will express -1,2% of interest.

2.28 Amount Details

Attribute	Type	Condition	Description
instructedAmount	Amount and Exchange Details	Optional	
transactionAmount	Amount and Exchange Details	Optional	

Example:

```
"amountDetails":
{
  "instructedAmount" :
  {
    "currency": "DKK",
    "amount": "123.50",
```

```

        "sourceCurrency": "DKK",
        "targetCurrency": "EUR",
        "exchangeRate": "7.47",
      },
      "transactionAmount" :
      {
        "currency": "EUR",
        "amount": "16.53",
        "sourceCurrency": "DKK",
        "targetCurrency": "EUR",
        "exchangeRate": "7.60",
      }
    }
  }

```

2.29 Amount and Exchange Details

Attribute	Type	Condition	Description
currency	Currency Code	Mandatory	ISO 4217 Alpha 3 currency code
amount	String	Mandatory	The amount given with fractional digits, where fractions must be compliant to the currency definition. Up to 14 significant figures. Negative amounts are signed by minus. The decimal separator is a dot. Example: Valid representations for EUR with up to two decimals are: • 1056 • 5768.2 • -1.50 • 5877.78
sourceCurrency	Currency Code	Optional	Currency from which an amount is to be converted in a currency conversion.
exchangeRate	String	Optional	Factor used to convert an amount from one currency into another. This reflects the price at which one currency was bought with another currency.

Attribute	Type	Condition	Description
			Note: If this field is present, then also the field sourceCurrency must be provided.
unitCurrency	Currency Code	Optional	Currency in which the rate of exchange is expressed in a currency exchange. In the example 1EUR = xxxCUR, the unit currency is EUR.
targetCurrency	Currency Code	Optional	Currency into which an amount is to be converted in a currency conversion.
quotationDate	ISODate	Optional	Date at which an exchange rate is quoted.
contractId	String	Optional	Unique identification to unambiguously identify the foreign exchange contract.

2.30 API Instance Parameters

Attribute	Type	Condition	Description
apiName	Max35Text	Mandatory	"Berlin Group openFinance", "giroAPI", "EPC SPAA", "Berlin Group XS2A", "Berlin Group Direct Access". More entries might be added by the API Provider.
releaseVersion	Max35Text	Mandatory	Release version, not necessary connected to the standard version.
validFrom	ISODate	Optional	
validTo	ISODate	Optional	
releaseNote	Max2000Text	Optional	Offers a summary on the new release.
_links	Links	Mandatory	A list of hyperlinks to be recognised by the TPP. The actual hyperlinks used in the response depend on the dynamical

Attribute	Type	Condition	Description
			decisions of the ASPSP when processing the request. Remark: All links can be relative or full links, to be decided by the ASPSP. The following valid values for links are supported: - aspspParameters - aspspContacts - aspspDowntimes

This data type is used in [oFA-Disc].

2.31 ASPSP API Access

Attribute	Type	Condition	Description
lastUpdate	ISODateTime	Mandatory	Last Update in one of the attributes of entities referenced by links provided in apiInstances
aspspName	Max70Text	Mandatory	Name ASPSP:
aspspBranch	Max70Text	Optional	Branch or Division in case of multiple APIs per ASPSP which relate to different branches/divisions
aspspChannel	Max70Text	Optional	Online Channel in case of multiple APIs per ASPSP Online Channel, e.g. "small corporates" or "retail".
aspspIdentifications	Array of BICFI	Mandatory	All bank identifications which are covered by the addressed API.
clearingSystemMemberId	Clearing System Member Identification	Optional	
accessScheme	Max35Text	Optional	Scheme-identification

Attribute	Type	Condition	Description
aspspDescription	Max140Text	Mandatory	Description of API provider party.
validFrom	ISODate	Optional	
validTo	ISODate	Optional	
contacts	Array of Contact	Optional	
_links	Links	Optional	A list of hyperlinks to be recognised by the API Client. The actual hyperlinks used in the response depend on the dynamical decisions of the ASPSP when processing the request. Remark: All links can be relative or full links, to be decided by the ASPSP. The following valid values for links are supported: - aspspContacts - apiInstances

This data type is used in [oFA-Disc].

2.32 Authentication Object

Attribute	Type	Condition	Description
authenticationType	Authentication Type	Mandatory	Type of the authentication method.
authenticationVersion	String	Conditional	Depending on the "authenticationType". This version can be used by differentiating authentication tools used within performing OTP generation in the same authentication type. This version can be referred to in the ASPSP's documentation.

Attribute	Type	Condition	Description
authenticationMethodId	Max35Text	Mandatory	An identification provided by the ASPSP for the later identification of the authentication method selection.
name	String	Mandatory	This is the name of the authentication method defined by the PSU in the Online Banking frontend of the ASPSP. Alternatively this could be a description provided by the ASPSP like "SMS OTP on phone +49160 xxxxx 28". This name shall be used by the TPP when presenting a list of authentication methods to the PSU, if available.
explanation	String	Optional	detailed information about the SCA method for the PSU

This data type is used in [oFA-Co], [oFA-IG-Com], [oFA-PFSM], [oFA-IG-EPIS] .

2.33 Authentication Type

More authentication types might be added during implementation projects and documented in the ASPSP documentation.

Name	Description
SMS_OTP	An SCA method, where an OTP linked to the transaction to be authorised is sent to the PSU through a SMS channel.
CHIP_OTP	An SCA method, where an OTP is generated by a chip card, e.g. an TOP derived from an EMV cryptogram. To contact the card, the PSU normally needs a (handheld) device. With this device, the PSU either reads the challenging data through a visual interface like flickering or the PSU types in the challenge through the device key pad. The

Name	Description
	device then derives an OTP from the challenge data and displays the OTP to the PSU.
PHOTO_OTP	An SCA method, where the challenge is a QR code or similar encoded visual data which can be read in by a consumer device or specific mobile app. The device resp. the specific app then derives an OTP from the visual challenge data and displays the OTP to the PSU.
PUSH_OTP	An OTP is pushed to a dedicated authentication APP and displayed to the PSU.
SMTP_OTP	An OTP is sent via email to the PSU.

2.34 Balance

Attribute	Type	Condition	Description
balanceAmount	Amount	Mandatory	
balanceType	Balance Type	Mandatory	
creditLimitIncluded	Boolean	Optional	A flag indicating if the credit limit of the corresponding account is included in the calculation of the balance, where applicable.
lastChangeDateTime	ISODateTime	Optional	This data element might be used to indicate e.g. with the expected or booked balance that no action is known on the account, which is not yet booked.
referenceDate	ISODate	Optional	indicates the date of the balance
lastCommitted Transaction	Max35Text	Optional	entryReference of the last committed transaction to support the TPP in identifying whether all PSU transactions are already known.

This data type is used in [oFA-IG-Com].

2.35 Binary Document

Base64 encoding of the binary file representing the enclosed document or template, such as a PDF file, image file, XML file, MT message. The maximum size of the binary document is 10MB.

2.36 Card Account Details

Attribute	Type	Condition	Description
resourceId	String	Mandatory	This is the data element to be used in the path when retrieving data from a dedicated account.
maskedPan	Max35Text	Mandatory	Primary Account Number (PAN) of the main card in masked form..
currency	Currency Code	Mandatory	Account currency
ownerName	Max140Text	Optional	Name of the legal account owner. If there is more than one owner, then e.g. two names might be noted here. For a corporate account, the corporate name is used for this attribute. Even if supported by the ASPSP, the provision of this field might depend on the fact whether an explicit consent to this specific additional account information has been given by the PSU.
name	Max70Text	Optional	Name of the account, as assigned by the ASPSP, in agreement with the account owner in order to provide an additional means of identification of the account.
displayName	Max70Text	Optional	Name of the account as defined by the PSU within online channels.

Attribute	Type	Condition	Description
product	Max35Text	Optional	Product Name of the Bank for this account, proprietary definition
debitAccounting	Boolean	Optional	If true, the amounts of debits on the reports are quoted positive with the related consequence for balances. If false, the amount of debits on the reports are quoted negative.
status	String	Optional	Account status. The value is one of the following: • "enabled": account is available • "deleted": account is terminated • "blocked": account is blocked e.g. for legal reasons If this field is not used, than the account is available in the sense of this specification.
usage	Max140 Text	Optional	Specifies the usage of the account - PRIV: private personal account - ORGA: professional account
details	Max1000 Text	Optional	Specifications that might be provided by the ASPSP - characteristics of the account - characteristics of the relevant card
creditLimit	Amount	Optional	Defines the credit limit of the PSU aggregated for all cards related to this card account in total.
balances	Array of Balances	Optional	The specific card account balances associated to this card accounts.
_links	Links	Optional	Links to the cardAccount, which can be directly used for retrieving account information from this dedicated account.

Attribute	Type	Condition	Description
			Links to "balances" and/or "cardTransactions" These links are only supported, when the corresponding consent has been already granted.

This data type is used in [oFA-IG-Com].

2.37 Card Account Report

Attribute	Type	Condition	Description
booked	Array of Card Transactions	Conditional	Shall be contained if bookingStatus parameter is set to "booked" or "both".
pending	Array of Card Transactions	Optional	Not contained if the bookingStatus parameter is set to "booked".
_links	Links	Mandatory	The following links might be used within this context: • cardAccount (mandatory when providing transaction reports on card reconciliation accounts under /card-accounts) • card (mandatory when providing transaction reports on single card entry level under /cards)first (optional) • next (optional) • previous (optional) • last (optional)

This data type is used in [oFA-IG-Com].

2.38 Card Transactions

Attribute	Type	Condition	Description
cardTransactionId	Max35Text	Optional	Unique end to end identity.
terminalId	Max35Text	Optional	Identification of the Terminal, where the card has been used.
transactionDate	ISODate	Optional	date of the actual card transaction
acceptorTransaction DateTime	ISODate Time	Optional	Timestamp of the actual card transaction within the acceptance system
bookingDate	ISODate	Optional	booking date of the related booking on the card account
valueDate	ISODate	Optional	Date at which assets become available to the account owner in case of a credit entry, or cease to be available to the account owner in case of a debit entry. For card transactions this is the payment due date of related booked transactions of a card.
transactionAmount	Amount	Mandatory	The amount of the transaction as billed to the card account.
grandTotalAmount	Amount	Optional	Total amount of the instalment including charges, insurance and taxes in addition to the funded amount.
currencyExchange	Array of Report Exchange Rate	Optional	For card accounts, this often is restricted by the ASPSP to use only one exchange rate.
originalAmount	Amount	Optional	Original amount of the transaction at the Point of Interaction in original currency

Attribute	Type	Condition	Description
markupFee	Amount	Optional	Any fee related to the transaction in billing currency.
markupFeePercentage	String	Optional	Percentage of the involved transaction fee in relation to the billing amount, e.g. "0.3" for 0,3%
cardAcceptorId	Max35Text	Optional	Identification of the Card Acceptor (e.g. merchant) as given in the related card transaction.
cardAcceptorName	Max70Text	Optional	Name of the Card Acceptor (e.g. merchant) as provided in the related card transaction
cardAcceptorAddress	Address	Optional	Address of the Card Acceptor as given in the related card transaction.
cardAcceptorPhone	Phone Number	Optional	Merchant phone number
merchantCategoryCode	Merchant Category Code	Optional	Merchant Category Code of the Card Acceptor as given in the related card transaction.
maskedPAN	Max35Text	Optional	The masked PAN of the card used in the transaction.
transactionDetails	Max1000Text	Optional	Additional details given for the related card transactions.
invoiced	Boolean	Optional	Flag indicating whether the underlying card transaction is already invoiced.
proprietaryBankTransactionCode	Max35Text	Optional	proprietary bank transaction code as used within a community or within an ASPSP e.g. for MT94x based transaction reports

2.39 Card Transaction

Attribute	Type	Condition	Description
card	Card Type	Optional	
transaction	Card Transaction Details	Optional	

2.40 Card Transaction Details

Attribute	Type	Condition	Description
paymentContext	Payment Context Type	Optional	
sequenceNumber	Max35Text	Optional	
transactionDateTime	ISODateTime	Optional	

2.41 Card Type

Attribute	Type	Condition	Description
cardBrand	Max35Text	Optional	
maskedPAN	Max30Text	Optional	
cardHolderName	Max45Text	Optional	

2.42 Challenge

Attribute	Type	Condition	Description
image	String	Optional	PNG data (max. 512 kilobyte) to be displayed to the PSU, Base64 encoding, cp. [RFC4648]. This attribute is used only, when PHOTO_OTP or CHIP_OTP is the selected SCA method.

Attribute	Type	Condition	Description
data	Array of Strings	Optional	A collection of challenge data
imageLink	String	Optional	A link where the ASPSP will provides the challenge image for the TPP.
otpMaxLength	Integer	Optional	The maximal length for the OTP to be typed in by the PSU.
otpFormat	String	Optional	The format type of the OTP to be typed in. The admitted values are "characters" or "integer".
additional Information	String	Optional	Additional explanation for the PSU to explain e.g. fallback mechanism for the chosen SCA method. The TPP is obliged to show this to the PSU.

This data type is used in [oFA-Co] ,[oFA-IG-Com], [oFA-PFSM], [oFA-IG-EPIS].

2.43 Clearing System Member Identification

Attribute	Type	Condition	Description
memberId	Max35Text	Optional	Identification of a member of a clearing system.
clearingSystemIdentificationCode	Clearing System Identification Code	{Or - Optional	Identification of a clearing system, in a coded form as published in an external list.
clearingSystemIdentificationProprietary	Max35Text	Or – Optional}	Identification code for a clearing system, that has not yet been identified in the list of clearing systems.

This data type is used in [oFA-Disc].

2.44 Client Message Information

Attribute	Type	Condition	Description
category	String	Mandatory	Only "ERROR" or "WARNING" permitted
code	Message Code	Mandatory	
path	String	Conditional	Path of the element of the request message which provoked the error message, if applicable
text	Max500Text	Optional	Additional explaining text.

This data type is used in [oFA-IG-Com], [oFA-PFSM], [oFA-IG-EPIS], [oFA-IG-RTP]

2.45 Contact

Attribute	Type	Condition	Description
contactType	Contact Type	Mandatory	
name	Max70Text	Optional	
postalAddress	Postal Address	Optional	
contactDetails	Contact Details	Optional	

This data type is used in [oFA-Disc].

2.46 Contact Details

Attribute	Type	Condition	Description
namePrefix	Name Prefix Code	Optional	Specifies the terms used to formally address a person.
name	Max70Text	Optional	Name by which a party is known and which is usually used to identify that party.
phoneNumber	Max35Text	Optional	pattern \+[0-9]{1,3}-[0-9()+\-]{1,30}
mobileNumber	Max35Text	Optional	pattern \+[0-9]{1,3}-[0-9()+\-]{1,30}

Attribute	Type	Condition	Description
faxNumber	Max35Text	Optional	pattern \+[0-9]{1,3}-[0-9()+\-]{1,30}
emailAddress	Max70Text	Optional	Address for electronic mail (e-mail).
emailPurpose	Max35Text	Optional	Purpose for which an email address may be used.
jobTitle	Max35Text	Optional	Title of the function.
responsibility	Max35Text	Optional	Role of a person in an organisation.
department	Max35Text	Optional	Identification of a division of a large organisation or building.
preferredMethod	Preferred Method	Optional	Preferred method used to reach the contact.

2.47 Creditor Details and Account

Attribute	Type	Condition	Description
creditor	Party Description	Mandatory	
creditorAccount	Account Reference	Mandatory	

This data type is used in [oFA-Co].

2.48 Creditor Reference Information

Attribute	Type	Condition	Description
reference	Max35Text	Mandatory	The actual reference.
referenceTypeCode	Creditor Reference Type Code	{Or Optional	Type of the reference, provided as code.
referenceTypeProprietary	Max35Text	Or Optional}	Type of the reference, provided in proprietary encoding.

Attribute	Type	Condition	Description
referenceIssuer	Max35Text	Optional	Issuer of the reference.

2.49 Direct Debit Transaction (Mandate Information)

Attribute	Type	Condition	Description
mandateRelatedInformation	Mandate Related Information	Mandatory	
creditorSchemeld	Party Description3	Optional	<u>SDD</u> : Identification = Creditor-Identifier (CI) "schemeNameProprietary" : "SEPA"

This data type is used in [oFA-PFDom].

2.50 Duration

Attribute	Type	Condition	Description
fromDate	ISODate	Mandatory	
toDate	ISODate	Optional	

2.51 Enclosed File

Attribute	Type	Condition	Description
typeCode	Document Type Code	{Or Optional	Type of document or template.
typeProprietary	Max35Text	Optional Or}	Proprietary document type definition.
identification	Max35Text	Mandatory	Identification of the document or template.
issueDate	ISODate	{Or	Issue date of the document.

Attribute	Type	Condition	Description
issueDateTime	ISODateTime	Or}	Issue date and time of the document.
name	Max140Text	Optional	Name of document or transaction, for example, tax invoice.
languageCode	Language Code	Optional	Unique identifier for a language used in the document.
formatCode	Document Format Code	{Or	Format of the document or template, such as PDF, XML, XSLT.
formatProprietary	Max35Text	Or}	Proprietary format encoding.
formatProprietary SchemeName	Max35Text	Optional	May be used only if also formatProprietary is used.
formatProprietary Issuer	Max35Text	Optional	May be used only if also formatProprietary is used.
fileName	Max140Text	Optional	Technical name of the file.
digitalSignature	Party And Signature	Optional	Digital signature of the enclosed binary file. Note: Not supported for now. Might be added later in case of market needs.
enclosure	Binary Document	Mandatory	Base64 encoding of the binary file representing the enclosed document or template, such as a PDF file, image file, XML file, MT message.

This data type is used in [oFA-PFDom].

2.52 Entry Details

This data type describe entry details of a batch booking entry. The details are restricted to data attributes relevant in this case.

Attribute	Type	Condition	Description
references	References	Optional	might contain the endToEndId as well as the new UETR field
creditor	Party Description	Optional	Name and potentially an identification of the creditor if a "Debited" transaction

Attribute	Type	Condition	Description
transactionAmount	Amount	Mandatory	The amount of the transaction as billed to the account.
currencyExchange	Array of Report Exchange Rate	Optional	
creditorAccount	Account Reference	Conditional	
creditorAgent	Agent Description1	Optional	Proprietary identifications might be provided under financialInstitutionId/other
ultimateCreditor	Party Description4	Optional	
debtor	Party Description	Optional	Name and potentially identification of the debtor if a "Credited" transaction
debtorAccount	Account Reference	Conditional	
debtorAgent	Agent Description1	Optional	Proprietary identifications might be provided under financialInstitutionId/other
ultimateDebtor	Party Description4	Optional	
remittanceInformationUnstructured	Array of Max140Text	Optional	
remittanceInformationStructured	Array of Re-mittance	Optional	
purposeCode	Purpose Code	Optional	

2.53 Error Information

This is a data element to support the declaration of additional errors in the context of [RFC7807].

Attribute	Type	Condition	Description
title	Max70Text	Optional	Short human readable description of error type. Could be in local language. To be provided by ASPSPs.
detail	Max500Text	Optional	Detailed human readable text specific to this instance of the error. XPath might be used to point to the issue generating the error in addition. Remark for Future: In future, a dedicated field might be introduced for the XPath.
code	Message Code	Mandatory	Message code to explain the nature of the underlying error.

This data type is used in [oFA-PFSM].

2.54 Evaluated Amount

This data structure represents an evaluated amount (without indicating the method of evaluation) when an unambiguous balance cannot be provided, but an evaluation, e.g. the total value of a securities deposit:

Attribute	Type	Condition	Description
amount	Amount	Mandatory	Amount which is the result of the corresponding evaluation.
evaluationDateTime	ISODateTime	{Or	Date / date and time of the evaluation.
evaluationDate	ISODate	Or}	

2.55 Evaluated Price

This data structure represents an evaluated price (e.g. of a security) when an unambiguous balance cannot be provided, but an evaluation, e.g. the total value of a securities deposit

Attribute	Type	Condition	Description
amount	Amount	{Or	If the price is evaluated as an amount, amount of the price in its respective currency.
percentage	String	Or}	Percentage of the price with (optional) fractional digits. Up to 20 significant figures. Negative amounts are signed by minus. The decimal separator is a dot. Trailing zeroes must be truncated.
priceDateTime	ISODateTime	{Or	Date / date and time of the evaluation of this price.
priceDate	ISODate	Or}	
priceType	Type of Price Code	Optional	Type of the price.
sourceOfPrice	Source of Price	Optional	Indicates the source of the respective price.
description	Max500Text	Optional	Additional description of the source of price. E.g., if the source of price indicates a local market, the ASPSP can include the local market's name in this field.
exchangeRates	Array of Report Exchange Rate	Optional	Relevant exchange rates for the determined price.

2.56 Extended Party Description

Attribute	Type	Condition	Description
name	Max140Text	Optional	Name of the party.

Attribute	Type	Condition	Description
postal address	Postal Address	Optional	Postal Address of the party.
identification	Party Identification	Optional	Identification of the party.
countryOf Residence	Country Code	Optional	Country in which a person resides (the place of a person's home). In the case of a company, it is the country from which the affairs of that company are directed.
contactDetails	Contact Details	Optional	Set of elements used to indicate how to contact the party.

2.57 Financial Institution Identification

Attribute	Type	Condition	Description
bicfi	BICFI	optional	Code allocated to a financial institution.
clearingSystemMemberId	Clearing System Member Identification	optional	Information used to identify a member within a clearing system.
name	Max140Text	optional	Name of the financial institution
postalAddress	Postal Address	Optional	Postal Address of the financial institution.
other	Generic Financial Institution Identification	Optional	Unique identification of an organisation, as assigned by an institution, using an identification scheme.

2.58 Financial Institution Identification1

Attribute	Type	Condition	Description
bicfi	BICFI	Mandatory	Code allocated to a financial institution.

2.59 Financial Instrument

This data structure represents financial instrument.

Attribute	Type	Condition	Description
isin	ISIN	{Or	ISIN of the financial Instrument. The primitive ISIN type is defined as in ISO20022.
other	Generic Financial Instrument Identification	Or}	If an ISIN is not available, the ASPSP may instead include an identification by other means.
name	Max70Text	Optional	Name of the financial instrument.
normalisedPrice	Evaluated Price	Optional	Price per unit of the financial instrument in case of a price amount. Percentage price otherwise. If used in the context of a transaction, the price here shall indicate the applied price (without fees) for the trade.

Remark: Publicly available information on the financial instrument like interests (e.g. in the case of bonds), various dates (maturity date, issue date,..) or details on options are intentionally not supported by the data model. Individual markets might extend the definition, for example to meet certain regulatory requirements.

2.60 Generic Account Identification

Attribute	Type	Condition	Description
identification	Max35Text	Mandatory	Unique and unambiguous identification of an account.
schemeName Code	Account Identification Code	{Or - Optional	An entry provided by an external ISO code list
schemeName Proprietary	Max35Text	Or – Optional}	A scheme name defined in a proprietary way.
issuer	Max35Text	Optional	Issuer of the identification

2.61 Generic Financial Instrument Identification

This data structure represents the identification of a financial instrument by means other than ISIN.

Attribute	Type	Condition	Description
identification	Max35Text	Mandatory	Identification of a security.
suffix	Max16Text	Optional	Identifies the suffix of the security identification.
typeCode	External Financial Instrument Identification Type Code	{Or	Unique and unambiguous identification source, as assigned via a pre-determined code list.
typeProprietary	Max35Text	Or}	Unique and unambiguous identification source using a proprietary identification type.

2.62 Generic Financial Institution Identification

Attribute	Type	Condition	Description
identification	Max35Text	Mandatory	Unique and unambiguous identification of a financial institution.
schemeName Code	Financial Institution Identification Code	{Or - Optional	An entry provided by an external ISO code list
schemeName Proprietary	Max35Text	Or – Optional}	A scheme name defined in a proprietary way.
issuer	Max35Text	Optional	Issuer of the identification

2.63 Generic Organisation Identification

Attribute	Type	Condition	Description
identification	Max35Text	Mandatory	Unique and unambiguous identification of an organisation.
schemeName Code	Organisation Identification Code	{Or - Optional	An entry provided by an external ISO code list
schemeName Proprietary	Max35Text	Or – Optional}	A scheme name defined in a proprietary way.
issuer	Max35Text	Optional	Issuer of the identification

2.64 Geo Location

Format using [RFC2426], i.e. GEO:<latitude>;< longitude >.

This data type is used in [oFA-PFSM].

2.65 href Type

Attribute	Type	Condition	Description
href	String	Mandatory	

2.66 Index

Attribute	Type	Condition	Description
index	Max35Text	Optional	Though arbitrary text is technically supported, it is recommended to use the following keywords: "CPI" – customer Price Index "FC" – foreign currency "other" – for others, not clearly specified index relations.
rootIndexValue	Max35Text	Optional	A number (represented as a string) giving the basic index value which is the basis for the loan contract.
exchangeRate	Amount dependent Rate	Optional	Intended for usage in case of index = "FC".
additional Information	Max500Text	Optional	Explanations e.g. to specify index = "other".

2.67 Initiation

Initiation provides informations about entries in an /initiations sub resource of a payment resource.

Attribute	Type	Condition	Description
initiationId	Max35Text	Mandatory	resource identification of the addressed payment initiation.

Attribute	Type	Condition	Description
instructedAmount	Amount	Mandatory	Instructed Amount of the addressed initiation.
transactionStatus	Transaction Status	Mandatory	Status of the addressed initiation.
_links	Links	Mandatory	links of href type "paymentInitiation"

This data type is contained in [oFA-IG-EPIS].

2.68 Interest

Attribute	Type	Condition	Description
type	Interest Type Code	Optional	Type of the interest. Currently, interests are distinguished between those with fixed rates ("FIXD") and those with rates derived from an index ("INDE").
relatedIndices	Array of Index	Optional	List of indices related to the interest rate.
rate	Array of Amount dependent Rate	Mandatory	An array of interest rates, which might apply to different amount ranges.
fromDateTime	ISODateTime	Optional	The date from which this interest rate is applicable. If no exact time is applicable, 00:00:00 is chosen as time definition.
toDateTime	ISODateTime	Optional	The date until this interest rate is applicable. If no exact time is applicable, 23:59:59 is chosen as time definition.
reason	Max35Text	Optional	Specifies the reason of the interest. Though arbitrary text is technically supported, it is recommended to use the following keywords with pre-defined meaning: - arrears :The addressed interest rate applies only in case of arrears.

Attribute	Type	Condition	Description
			- alternateRateMaximumOf: The addressed alternate interest rate applies in case where this rate implies a higher interest in a dedicated interest period than the regular interest rate provided without reason element. - alternateRateMinimumOf: The addressed alternate interest rate applies in cases where this rate implies a lower interest in a dedicated interest period than the regular interest rate provided without reason element. ASPSPs may define further keywords and shall publish them in their documentation.
change Mechanism	Max1024Text	Optional	An explanation of a mandated change mechanism.

2.69 Lean Account Report

Attribute	Type	Condition	Description
booked	Array of transactions	Conditional	booked entries on the related account
pending	Array of transactions	Optional	pending entries on the related account
information	Array of transactions	Optional	information entries related to the addressed account.

This data type is used in [oFA PAIS].

2.70 Limit

Attribute	Type	Condition	Description
limitType	Limit Type	Mandatory	
limitAmount	Amount	Mandatory	

2.71 Links

The structure of Links is conform to [HAL].

Attribute	Type	Condition	Description
scaRedirect	href Type	Optional	A link to an ASPSP site where SCA is performed within the Redirect SCA approach.
scaOAuth	href Type	Optional	The link refers to a JSON document specifying the OAuth details of the ASPSP's authorisation server. JSON document 480llowst he definition given in [RFC 8414].
confirmation	href Type	Optional	"confirmation": Might be added by the ASPSP if either the "scaRedirect" or "scaOAuth" hyperlink is returned in the same response message. This hyperlink defines the URL to the resource which needs to be updated with • a confirmation code as retrieved after the plain redirect authentication process with the ASPSP authentication server or • an access token as retrieved by submitting an authorization code after the integrated OAuth based authentication process with the ASPSP authentication server.
startAuthorisation	href Type	Optional	A link to an endpoint, where the authorisation of a transaction or the authorisation of a transaction cancellation shall be started with a POST command. No specific data is needed for this process start.
startAuthorisationWithPsuIdentification	href Type	Optional	The link to an endpoint where the authorisation of a transaction or of a transaction cancellation shall be started, where PSU identification shall be uploaded with the corresponding call.
updatePsuIdentification	href Type	Optional	The link to the payment initiation or account information resource, which needs to be updated by the PSU identification if not delivered yet.

Attribute	Type	Condition	Description
startAuthorisationWithProprietaryData	hrefType	Optional	A link to the endpoint, where the authorisation of a transaction or of a transaction cancellation shall be started, and where proprietary data needs to be updated with this call. The TPP can find the scope of missing proprietary data in the ASPSP documentation. The usage of this hyperlink is not further specified in the specification but is used analogously to e.g. the startAuthorisation withPsuIdentification hyperlink.
updateProprietaryData	href Type	Optional	The link to the payment initiation or account information resource, which needs to be updated by the proprietary data.
startAuthorisationWithPsuAuthentication	href Type	Optional	The link to an endpoint where the authorisation of a transaction or of a transaction cancellation shall be started, where PSU authentication data shall be uploaded with the corresponding call.
updatePsuAuthentication	href Type	Optional	The link to the payment initiation or account information resource, which needs to be updated by a PSU password and eventually the PSU identification if not delivered yet.
startAuthorisationWithEncryptedPsuAuthentication	href Type	Optional	The link to an endpoint where the authorisation of a transaction or of a transaction cancellation shall be started, where encrypted PSU authentication data shall be uploaded with the corresponding call.
updateEncryptedPsuAuthentication	href Type	Optional	The link to the payment initiation or account information resource, which needs to be updated by an encrypted PSU password and eventually the PSU identification if not delivered yet.
startAuthorisationWithAuthenticationMethodSelection	href Type	Optional	This is a link to an endpoint where the authorisation of a transaction or of a transaction cancellation shall be started, where the selected SCA method shall be uploaded with the corresponding call.

Attribute	Type	Condition	Description
selectAuthenticationMethod	href Type	Optional	This is a link to a resource, where the TPP can select the applicable second factor authentication methods for the PSU, if there were several available authentication methods.
startAuthorisationWithTransactionAuthorisation	href Type	Optional	A link to an endpoint, where an authorisation of a transaction or a cancellation can be started, and where the response data for the challenge is uploaded in the same call for the transaction authorisation or transaction cancellation at the same time in the Embedded SCA Approach.
authoriseTransaction	href Type	Optional	The link to the payment initiation or consent resource, where the "Transaction Authorisation"Request" is sent to. This is the link to the resource which will authorise the payment or the consent by checking the SCA authentication data within the Embedded SCA approach.
updateResourceByDebtorAccountResource	href Type	Optional	A link to a payment related resource which needs to be updated by an account, referenced by a resource identification.
self	href Type	Optional	The link to the payment initiation resource created by the request itself. This link can be used later to retrieve the transaction status of the payment initiation.
status	href Type	Optional	A link to retrieve the status of the transaction resource.
transactionfees	hrefType	Optional	The link is to the status resource. This link is only added within the authorisation process in case fee information is available via the status resource.
scaStatus	href Type	Optional	A link to retrieve the status of the authorisation or cancellation-authorisation sub-resource.

Attribute	Type	Condition	Description
account	href Type	Optional	A link to the resource providing the details of one account
savingsAccount	href Type	Optional	A link to the resource providing the details of a savings account.
loanAccount	href Type	Optional	A link to the resource providing the details of a loan account.
balances	href Type	Optional	A link to the resource providing the balance of a dedicated account.
transactions	href Type	Optional	A link to the resource providing the transaction history of a dedicated account.
cardAccount	href Type	Optional	A link to the resource providing the details of one card account.
cardTransactions	href Type	Optional	A link to the resource providing the transaction history of a dedicated card account.
transactionDetails	href Type	Optional	A link to the resource providing details of a dedicated transaction.
ibanCheck	href Type	Optional	A link to the endpoint offering the addressed iban check result.
paymentInitiation	href Type	Optional	A link to an initiation related to a payment resource.
securitiesAccount	href Type	Optional	A link to the resource providing the details of one securitiesAccount.
positions	href Type	Optional	A link to the resource providing the list of positions of one securitiesAccount.
orders	href Type	Optional	A link to the resource providing the list of orders of one securitiesAccount.
orderDetails	href Type	Optional	A link to the resource providing details of one specific order.

Attribute	Type	Condition	Description
relatedOrders	Array of href Type	Optional	An array of links to request order details of related orders related to the order that is represented by the data structure containing this element (e.g. other orders that originated from the same order split as this order).
relatedTransactions	Array of href Type	Optional	An array of links to request securities transaction details of transactions that resulted from this order.
subscription	href Type	Optional	A link to the resource providing the details of a subscription for Push AIS Services.
entryStatusRevoked	Array of href Type	Optional	Links to entry endpoints where the entry status is revoked.
first	href Type	Optional	Navigation link for paginated account reports.
next	href Type	Optional	Navigation link for paginated account reports.
previous	href Type	Optional	Navigation link for paginated account reports.
last	href Type	Optional	Navigation link for paginated account reports.
download	href Type	Optional	Download link for huge AIS data packages.
paymentInitiation	href Type	Optional	link to an initiation related to a payment resource.
confirmInitiation	href Type	Optional	link to a confirmation endpoint, where a payment initiation needs to be confirmed explicitly by the API Client for execution.

This data type is part of the API Framework definition, introduced in [oFA-PFSM], but used in all Implementation Guidelines (not further specified here).

2.72 Mandate Related Information

Attribute	Type	Condition	Description
mandateId	Max35Text	Mandatory	
dateOfSignature	ISO Date	Mandatory	
amendmentInformationDetails	Amendment Information Details	Optional	
electronicSignature	Max1025Text	Optional	Only used for e-mandates

2.73 Mandate Type

Attribute	Type	Condition	Description
localInstrumentCode	Local Instrument Code	{Or	External List e.g. Core, B2B, DDMC (for electronic mandate)
localInstrumentProprietary	Max35Text	Or}	
serviceLevelCode	Service Level Code	{Or Optional	External code
serviceLevelProprietary	Max35Text	Or Optional}	
categoryPurposeCode	Category Purpose	{Or Optional	External List
categoryPurposeProprietary	Max35Text	Or Optional}	
classificationCode	Classification Code	{Or Optional	FIXE, USGB, VARI (internal code)
classificationProprietary	Max35Text	Or Optional}	

2.74 Mandate Status

Code	Description
received	The mandate data have been received and are technically correct. The data is not authorised yet.
rejected	The mandate data have been rejected e.g. since no successful authorisation has taken place.
partiallyAuthorised	The mandate is due to a multi-level authorisation, some but not all mandated authorisations have been performed yet.
valid	The mandate is accepted and valid for direct debit transactions specified in the mandate object.
revokedByPsu	The mandate has been revoked by the PSU towards the ASPSP.
expired	The mandate expired.
terminatedByTpp	The corresponding TPP has terminated the mandate by applying the DELETE method to the mandate resource.

The ASPSP might add further codes. These codes then shall be contained in the ASPSP's documentation of the openFinance API.

2.75 Market Identification

Attribute	Type	Condition	Description
mic	Max4Text	{Or	ISO 10383 code of the market place.
market Identifier Proprietary	Max35Text	Or}	Proprietary identifier of the market place.

2.76 Occurrences

Attribute	Type	Condition	Description
sequenceType	Sequence Type	Mandatory	RCUR, OOF
frequencyType	Frequency Code	Optional	
duration	Duration	Optional	Length of time for which the mandate remains valid.
firstCollectionDate	ISODate	Optional	Date of the first collection of a direct debit as per the mandate.
finalCollectionDate	ISODate	Optional	Date of the final collection of a direct debit as per the mandate.

2.77 Organisation Identification

Attribute	Type	Condition	Description
anyBIC	BIC	{Or	A code allocated to a business entity or to a financial institution by a Registration Authority under an international identification scheme.
lei	LEI	Or	Legal Entity Identifier.
others	Array of Generic Organisation Identification	Or}	Unique identification of an organisation, as assigned by an institution, using an identification scheme.

2.78 Parameters API Security

Attribute	Type	Condition	Description
messageSigningRequired	Boolean	Mandatory	Mandate the API Client to sign the requests on application level. • True: The fields x-jws-signature and Digest have to be included in the message header. In addition, the TLS Certificate is used on transport layer. • False: Only the TLS Certificate on the transport layer is used. The ASPSP has to ensure that a check of access rights can be performed. This check has to be done previous to changing the transport layer and access right information which are contained in the certificate being potentially removed, e.g. by firewalls. (IO1)
messageSigningCertificate Mandated	Boolean	Mandatory	Indicates whether the ASPSP mandates the API Client to include the certificate used for message signing into the message (as part of the JWS Protected Header).
supportedMessageSignature Algorithms	Array of Signature Algorithm Code	Conditional	Array [] containing one or more identifier indicating signature algorithms according to RFC7518, i.e. RS256, PS512, etc. applicable to http message signing.
scaMethods	Array of Authentication Type	Mandatory	Supported SCA Methods (IO11)
scaApproaches	Array of SCA Approach Type	Mandatory	One or several of the following values: • Decoupled • Embedded • OAuth2 • Redirect (IO5)

Attribute	Type	Condition	Description
scaApproaches MethodsMapping	SCA Approach Methods Mapping	Optional	Configuration of supported SCA methods per applicable SCA Approaches • Decoupled • Embedded • Oauth2 • Redirect (IO12)
oauth2PrestepRequired	Boolean	Mandatory	Oauth2 required as a pre-step for PSU authentication (of the first factor) The ASPSP faces the following options: • True: The ASPSP requires a Oauth2 prestep before sending the first request to the openFinance or XS2A API. The API Client has to perform an Oauth2 based authorization prior to the openFinance service request. This authorisation covers only the first factor of an SCA of the full (simple) authorisation if no second factor is used, e.g. in case of exemptions. If an Oauth2 prestep is used, the field Authorization in the header has to carry the bearer token Oauth2 authorization. • False: No Oauth2 prestep is used. (IO6)
oauth2PrestepMetadataUri	Max140Text	Optional	The URI of the Oauth implementation Metadata information.
oauth2OfflineScopeRequired	Boolean	Optional	Defines, whether the Oauth scope "offline_access" needs to be supported when addressing the API where the PSU is not online.

Attribute	Type	Condition	Description
scaProcessFlowRequirement	SCA Process Flow Requirement Code	Optional	There are some options regarding the SCA process flow: there are various points in the process flow where the selection of the SCA method could be performed. (IO9)
client-explicit-authorisation-preferredSupported	Boolean	Mandatory	Indicates whether the ASPSP will not start the authorisation implicitly, if Client-Explicit-Authorisation-Preferred equals true.
supportedStartAuthorisationHyperlinks	Array of Max35Text	Mandatory	All "startAuthorisationXXX" hyperlinks supported in this API. If the Array is empty, start authorisation is not supported in this API instance. (IO39)
supportedUpdateAuthorisationHyperlinks	Array of Max35Text	Mandatory	All "updateAuthorisationXXX" hyperlinks supported in this API. If the Array is empty, update authorisation is not supported in this API instance.
firstFactorViaApiForDecoupledScaRequired	Boolean	Mandatory	In case of a decoupled SCA approach, parts of the PSU authentication might still be required via the API. There are essentially two variants for the implementation: False i.e. PSSU Identification sufficient: Only the PSU-Id is transmitted via the API. Information beyond that are not transferred via the API. True: (OF-)Authentication required: The first factor of the SCA is handled via the API. The remaining part of the SCA is not processed via the API.

Attribute	Type	Condition	Description
			(IO14)
supportedBodySignatureAlgorithms	Array of Signature Algorithm Code	Conditional	Array [] containing one or more identifier indicating signature algorithms according to RFC7518, i.e. RS256, PS512, etc. applicable to body signing algorithms.
passwordEncryptionRequired	Boolean	Mandatory	PSU Authentication data delivered via the XS2A-Interface (Embedded Approach) shall be encrypted at application level. (IO40)
supportedKeyEncryptionAlgorithms	Array of Key Encryption Algorithm Code	Conditional	Array [] containing one or more identifier indicating encryption algorithms for key encryption according to RFC7518, i.e. RSAES-OAEP, RSA-OAEP-256, etc.
supportedContentEncryptionAlgorithms	Array of Content Encryption Algorithm Code	Conditional	Array [] containing one or more identifier indicating encryption algorithms for encryption of the content according to RFC7518, i.e. A256GCM, etc.
implicitTransactionAuthorisationSupported	Boolean	Mandatory	Implicit start of transaction authorisation supported. (IO38)
psuNameInAuthorisationSupported	Boolean	Mandatory	Indicates, whether providing of the PSU Name in the authorisation sub resource is supported. (IO68)
supportedTppScaPreferences	Array of Max35Text	Mandatory	Indicates all attributes for SCA preferences which are supported by the ASPSP. Potential values: "TPP-Decoupled-Preferred", "TPP-Redirect-Preferred" and "TPP-SCA-Approach-Preference" If array is empty,

Attribute	Type	Condition	Description
			no SCA preference choice by the TPP is supported by the ASPSP. (IO71)

2.79 Parameters Payment Service

Attribute	Type	Condition	Description
serviceType	Service Type	Mandatory	The addressed payment initiation service, e.g. "XMDPIS", where the configuration data below relates to.
serviceVersion	Max10Text	Mandatory	Version of Implementation Guidelines of the addressed service, e.g. "2.1".
statusNotificationSupported	Boolean	Mandatory	Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources.
psuHeaderData	PSU Header Data	Mandatory	Defines whether PSU ID, Type etc are mandated by the ASPSP in the first transaction request.
confirmationRequired	Boolean	Mandatory	The confirmation step in authorization Redirect SCA or OAuth2 SCA Approach required for this service. (IO57)
multiLevelScaSupported	Boolean	Mandatory	Indicates, whether the service is offered as a service mandating a multi-level authorisation for some PSU. (IO27)
scaForCancellationRequired	Boolean	Conditional	It is optional to require an SCA to cancel the related payment service.

Attribute	Type	Condition	Description
			• True: SCA can be required for payment cancellations where applicable. The ASPSP is free to drop the SCA in some cases. • False: SCA is not required to cancel payments of the related service. Attribute not provided: Cancellation of the related service is not supported. (IO26)
supportedStatusFormats	Array of Max35Text	Mandatory	Supported codes: "JSON", "pain.002.0x.0x", where x is instantiated with the correct values. (IO44)
psuNameInStatusSupported	Boolean	Mandatory	Indicates whether the psuName is provided in resource status. (IO67)
jsonPaymentProducts	Array of Payment Product JSON	Optional	List of supported JSON based payment products. with related product specific additional information. (IO2, IO3, IO4)
xmlPaymentProducts	Array of Payment Product XML	Optional	List of supported XML based payment products.
textPaymentProducts	Array of Payment Product Text	Optional	List of supported text based payment products.
debtorNameReturned	Boolean	Mandatory	Indicates whether the ASPSP will return the debtor name in JSON/XML

Attribute	Type	Condition	Description
			based payment initiations after successful SCA. (IO58)
fundsAvailableSupported	Boolean	Optional	This attribute indicates whether the attribute "fundsAvailable" is supported in payment status.
transactionFeesSupported	Boolean	Mandatory	Transaction fees transported via the openFinance API in Payment Initiation Services. (IO10)
supportedScaExemptions	Array of SCA Exemption Code	[0..*]	Exemptions from SCA are possible in general. An ASPSP has to decide whether and if so under which condition to allow for an exemption from SCA for payment initiation. The applied condition is arbitrary but it has to be exactly one condition that determines the scenario for an SCA exemption. Empty array implies that SCA exemption is not supported. No entry of this field implies no information. (IO22)
minLead	Integer	Optional	The minimum of bank working days from the current date which is needed for the requested execution date.
maxLead	Integer	Optional	The maximum of calendar days from the current date which is supported for the requested execution date.
maximumExpiryDays Reservation	Integer	Optional	The maximum expiry interval for a reservation of funds.

Attribute	Type	Condition	Description
supportedExecutionPeriods	Array of Frequency Code	Optional	Supported frequency codes for recurring payments.
supportedMaximumExecutionDay	Integer	Optional	Only relevant for recurring payments. Either 30 or 31 as possible values. If 30 is the maximum execution day in a month, then "ultimo" is not supported as value.
supportedExecutionRules	Array of Max35Text	Optional	The supported values are "preceding" and "following".
supportedAccountReference	Array of Account Reference Code	Mandatory	List of account reference codes which are supported for the debtor account. (IO18)
accountCurrencyCode Required	Boolean	Mandatory	Defines whether the PSU account identification mandates the currency code for identification.
multiCurrencyAccounts Supported	Boolean	Mandatory	The ASPSP supports multi-currency accounts for payment initiation services. (IO17)
supportedAdditionalCharacterSetTextfields	Array of Max1Text	Mandatory	Characters in addition to the standard character set as defined in [oFA-PFSM], which are accepted in the payment API for text fields (names, address and remittance information). If the array is empty, no additional characters are permitted in the API.
transformedAdditionalCharacterSetTextfields	Array of Max1Text	Optional	Additional Characters accepted for text fields within payments (see above), which are transformed into a different character/characters in addition.

Attribute	Type	Condition	Description
supportedCountryCodes	Array of Country Codes	Optional	List of country codes for cross-border payment products.
supportedCurrencyCodes	Array of Currency Code	Optional	List of currencies which are accepted for cross-border payments. (IO52)
supportedInstructionCodes	Array of Instruction Code	Optional	Specifies further instructions concerning the processing of a payment instruction, provided by the sending clearing agent to the next agent(s).
batchBookingFlagSupported	Boolean	Optional	if equals true, then the ASPSP will book the bulk transactions following this flag.
maximumNumberTransactions	Integer	Optional	maximum number of transactions in a bulk payment
totalAmountRequired	Boolean	Optional	Applicable only for bulk payments
_links	Links	Mandatory	The endpoint of the addressed payment service.

This data type is used in [oFA-Disc].

2.80 Parameters Account Information Service

Attribute	Type	Condition	Description
serviceType	Service Type	Mandatory	The addressed account information service, e.g. "AIS-PAY" "AIS-CAC", where the configuration data below relates to.
supportedAccessRightCodes	Array of Access Right Code	Mandatory	Array of access right codes which might be used when submitting a consent to an account information

Attribute	Type	Condition	Description
			service related to the addressed AIS service. (IO30,54,56)
supportedReportFormats	Array of Max35Text	Optional	supported scheme names for XML based payment initiation, e.g. "camt.05x.0x.0x" as well as "JSON" and "MT94x". (IO31)
globalStatementsSupported	Boolean	Mandatory	Defines whether statements on a dedicated endpoint /statements as introduced in [oFA-TDM2] are supported within this service type or not.
supportedEndpoints	Array of Max35Text	Mandatory	"accountDetails", "balances", "transactions", "statements", "orders" as permitted values for the array entries. (IO59, 61)
supportedQueryParameters Balances	Array of Max35Text	Mandatory	Query parameters supported for balances. "dateFrom" as potential value. (IO66)
supportedQueryParameters Transactions	Array of Max35Text	Mandatory	Query parameters supported for transactions. "dateFrom", "dateTo", "entryReferenceFrom", "bookingStatus", "deltaList", "withBalance", "pageSize" as potential values (IO32,33)

Attribute	Type	Condition	Description
maxPageSize	Integer	Conditional	Delivers the maximum page size in numbers of transactions to be reported. To be provided if the pageSize query parameter is supported.
supportedBookingStatusCode	Array of Max35Text	Conditional	IF the bookingStatus is supported for the addressed service, this attribute is mandated. It provides all booking status codes supported as query parameters. ""booked", "pending", "both", "information", "all" as potential values.
supportedQueryParameters Accounts	Array of Max35Text	Mandatory	Query parameters supported for account list/details. Potential values are "withBalance" and "evaluationCurrency". (IO32)
multiCurrencyAccountDetails	Max35Text	Optional	"aggregation", "sub-account" "aggregationAndSub-account" for potential solutions offered for account details in case of multi-currency accounts (IO41)
supportedInterestReasons	Array of Max35Text	Conditional	Shows all supported values for interest reasons defined for the data type Interest for account types with an interest definition. Shall be used in case the reason attribute is used at all.
countingOfAisRequests	Max35Text	Mandatory	This attribute is defining the way of counting a 4 times a day counting for PSU not present (cp. [EBA-RTS]) "none": no dedicated AIS request counting is performed for this service.

Attribute	Type	Condition	Description
			"timeslot": ,Counting is done by aggregating all related account information endpoint accesses in a certain time interval as one access. "byEndpointCall": Counting is done by allowing for each endpoint 4 times access per day, counting them explicitly by endpoint. (IO47)
maximumInformationInterval	Integer	Mandatory	Maximum addressable transaction information of the past, interval in days. Might be further restricted on account level.
supportedBalanceTypes	Array of Balance Type	Mandatory	Balance Types supported. (IO34,64,65)
creditLineIncludedSupported	Boolean	Mandatory	
standingOrdersSupported	Boolean	Mandatory	
supportedPageSize	Integer	Optional	
fileDownloadRequired	Boolean	Mandatory	Transaction list offered only as download link in case of too high volumes. (IO36,35)
signedAccountInformation Supported	Boolean	Mandatory	Signature on body level as defined in [oFA-PFSM].
supportedSignature Algorithms	Array of Signature Algorithm Code	Conditional	Array [] containing one or more identifier indicating signature algorithms according to RFC7518, i.e. RS256, PS512, etc.
encryptedAccountInformation Supported	Boolean	Mandatory	Encryption on application level supported by the ASPSP as defined

Attribute	Type	Condition	Description
			in [oFA-PFSM] and if a related certificate has been provided by the API Client.
_links	Links	Mandatory	The endpoint of the addressed account information service, e.g. of the related "/accounts" endpoint.

This data type is used in [oFA-Disc].

2.81 Parameters Consent Service

Attribute	Type	Condition	Description
serviceType	Service Type	Mandatory	The addressed account information service, e.g. "CONS-AIS"
statusNotificationSupported	Boolean	Mandatory	Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources.
psuHeaderData	PSU Header Data	Mandatory	Defines whether PSU ID, Type etc are mandated by the ASPSP in the first transaction request.
confirmationRequired	Boolean	Mandatory	The confirmation step in authorization Redirect SCA or OAuth2 SCA Approach required for this service. (IO57)
multiLevelScaSupported	Boolean	Mandatory	Indicates, whether the service is offered as a service mandating a multi-level authorisation for some PSU. (IO27)
inconsistentConsentRejected	Boolean	Optional	Indicates, whether an inconsistent consent is rejected before PSU authentication, e.g. where the PSU ID does not match the required IBAN. (IO53)
supportedAccountReference	Array of Account	Mandatory	List of account reference codes which are supported for the account in the

Attribute	Type	Condition	Description
	Reference Code		related account category in consents of type "detailed". (IO60,62)
psuNameInResourceSupported	Boolean	Optional	Indicates whether the psuName is provided in resource status. (Only relevant for Version 1.3.x). (IO67)
psuNameInStatusSupported	Boolean	Mandatory	Indicates whether the psuName is provided in authorisation status.
frequencyPerDay	Integer	Conditional	The maximum frequency offered for access without PSU being online. Only relevant for account information consent.
consentValidityDays	Integer	Optional	Maximum consent validity days without the validity being restricted by the ASPSP, if applicable.
supportedConsentTypes	Array of Consent Type Code	Mandatory	
multiCurrencyAccounts Supported	Boolean	Mandatory	The ASPSP supports multi-currency accounts in the consent API. (IO17)
revocationViaAspspSupported	Boolean	Mandatory	Indicates whether the ASPSP supports the revocation of a consent by the PSU via the ASPSP. (IO51)
supportedAccessRight Codes	Array of Access Right Code	Optional	Array of access right code admitted for the related consent, if it is not a consent on AIS. In the latter case, the supported access right codes are specific to the account type, which is why it is defined within the AIS information.

Attribute	Type	Condition	Description
_links	Links	Mandatory	The endpoint of the addressed consent service e.g. for "/consents/account-information"

This data type is used in [oFA-Disc].

2.82 Parameters Funds Confirmation Service

Attribute	Type	Condition	Description
serviceType	Service Type	Mandatory	The addressed funds confirmation service "XFCON"
cardNumberSupported	Boolean	Mandatory	Indicates whether the card number can be used for addressing the account in addition. If true and if the addressed account is a multicurrency account, then the card will address the related sub-account of the consent process, where the card number has been used.

This data type is used in [oFA-Disc].

2.83 Parameters Push AIS Service

Attribute	Type	Condition	Description
serviceType	Service Type	Mandatory	The addressed Push AIS sub service, "XAIPS-XX[X]", e.g. "XAIPS-RTP".
statusNotificationSupported	Boolean	Mandatory	Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources.
psuHeaderData	PSU Header Data	Mandatory	Defines whether PSU ID, Type etc are mandated by the ASPSP in the first transaction request.
confirmationRequired	Boolean	Mandatory	The confirmation step in authorization Redirect SCA or OAuth2 SCA Approach required for this service.

Attribute	Type	Condition	Description
			(IO57)
multiLevelScaSupported	Boolean	Mandatory	Indicates, whether the service is offered as a service mandating a multi-level authorisation for some PSU. (IO27)
psuNameInStatusSupported	Boolean	Mandatory	Indicates whether the psuName is provided in authorisation status. (IO67)
apiClientSecondaryPush URISupported	Boolean	Mandatory	Support of secondary Push URI registration
staticCallbackTextSupported	Boolean	Mandatory	Pushing callback text supported.
callbackLinksSupported	Boolean	Mandatory	Pushing links to related account information entries supported.
pushEntriesSupported	Boolean	Mandatory	Pushing actual account information supported.
attributeFilteringSupported	Boolean	Mandatory	Support of attribute filters when pushing
documentForwardingSupported	Boolean	Mandatory	Support of pushing also documents related to the account information pushed.
supportedAccountEntry Criteria	Array of Max35Text	Optional	The list of supported Account Entry Trigger Criteria
supportedAccountStatement Criteria	Array of Max35Text	Optional	The list of supported Account Statement Trigger Criteria
supportedAccountBalance Criteria	Array of Max35Text	Optional	The list of supported Account Balance Trigger Criteria
supportedRtpCriteria	Array of Max35Text	Optional	The list of supported RTP Trigger Criteria

Attribute	Type	Condition	Description
_links	Links	Mandatory	The endpoint of the addressed subscription service.

This data type is used in [oFA-Disc].

2.84 Parameters Signing Basket Service

Attribute	Type	Condition	Description
serviceType	Service Type	Mandatory	The addressed signing basket function "SGNB"
statusNotificationSupported	Boolean	Mandatory	Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources.
psuHeaderData	PSU Header Data	Mandatory	Defines whether PSU ID, Type etc are mandated by the ASPSP in the first transaction request.
confirmationRequired	Boolean	Mandatory	The confirmation step in authorization Redirect SCA or OAuth2 SCA Approach required for this service. (IO57)
multiLevelSCASupported	Boolean	Mandatory	Indicates, whether the service is offered as a service mandating a multi-level authorisation for some PSU. (IO27)
transactionGroupings	Array of Transaction Type	Mandatory	The maximum groups of transaction which might be grouped for authorisation. The constant "all" is indicating that there are no restrictions. (IO25)

This data type is used in [oFA-Disc].

2.85 Parameters RTP Service

Attribute	Type	Condition	Description
serviceType	Service Type	Mandatory	The addressed consent service, e.g. "XRTP"
statusNotificationSupported	Boolean	Mandatory	Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources.
documentSupported	Boolean	Mandatory	Expresses, whether the transport of a document within the submitted RTP is supported.
jsonPaymentProducts	Array of Payment Product JSON	Mandatory	List of supported JSON based payment products. with related product specific additional information.
_links	Links	Mandatory	The endpoint of the addressed rtp service.

This data type is used in [oFA-Disc].

2.86 .Party Description

Attribute	Type	Condition	Description
name	Max140Text	Optional	Name of the party.
postal address	Postal Address	Optional	Postal Address of the party.
identification	Party Identification	Optional	Identification of the party.
additionalPartyInformation	Additional Party Information	Optional	Only supported in extended services for creditor related parties, if at all. NOTE: This information will be neither put to the related interbank payment nor to account reports/statements and might only be used to

Attribute	Type	Condition	Description
			display related information to the PSU during potential authorisation processes.

This data type is used in [oFA-PFDom].

2.87 Party Description1

Attribute	Type	Condition	Description
name	Max140Text	Optional	Name of the party.

This data type is used in [oFA-PFDom].

2.88 Party Description2

Attribute	Type	Condition	Description
name	Max140Text	Optional	Name of the party.
postal address	Postal Address	Optional	Postal Address of the party.

This data type is used in [oFA-PFDom].

2.89 Party Description3

Attribute	Type	Condition	Description
identification	Party Identification1	Optional	Identification of the party.

This data type is used in [oFA-PFDom], [oFA-IG-Adm].

2.90 Party Description4

Attribute	Type	Condition	Description
name	Max140Text	Optional	Name of the party.
identification	Party Identification	Optional	Identification of the party.

2.91 Party Identification

Attribute	Type	Condition	Description
organisationId	Organisation Identification	{Or	An entry provided by an external ISO code list
privateId	Private Identification	Or }	A scheme name defined in a proprietary way.

Note: Only one of the entries is allowed.

This data type is used in [oFA PAIS].

2.92 Party Identification1

Attribute	Type	Condition	Description
privateId	Private Identification	Mandatory	A scheme name defined in a proprietary way.

2.93 Payment Conditions

Attribute	Type	Condition	Description
amountModificationAllowed	Boolean	optional	Indicates if the debtor is allowed to pay a different amount then the requested amount.
earlyPaymentAllowed	Boolean	optional	Indicates if the debtor is allowed to pay before the requested execution date.

2.94 Payment Context Type

Attribute	Type	Condition	Description
cardPresent	Boolean	Optional	
cardDataEntryMode	Card Data Entry Mode	Optional	

2.95 Payment Exchange Rate

Attribute	Type	Condition	Description
unitCurrency	Currency Code	Optional	Currency in which the rate of exchange is expressed in a currency exchange. In the example 1EUR = xxxCUR, the unit currency is EUR.
exchangeRate	String	Optional	Factor used to convert an amount from one currency into another. This reflects the price at which one currency was bought with another currency.
contractIdentification	Max35Text	Optional	Unique identification to unambiguously identify the foreign exchange contract.
rateType	Exchange Rate Type Code	Optional	Specifies the type used to complete the currency exchange. Only SPOT, SALE and AGRD is allowed.

This data type is used in [oFA-PFDom].

2.96 Payment Identification

Attribute	Type	Condition	Description
instructionId	Max35Text	Optional	Unique identification as assigned by an instructing party for an instructed party to unambiguously identify the instruction.

Attribute	Type	Condition	Description
endToEndId	Max35Text	Optional	
uetr	UUID	Optional	Universally unique identifier to provide an end-to-end reference of a payment transaction. It is recommended to use time-based version (version 1) and variant 8,9,a or b of the UUID as defined in [RFC4122] for UUID fields such as uetr.

This data type is used in [oFA-PFDom], [oFA-IG-EPIS].

2.97 Payment Identification1

Attribute	Type	Condition	Description
endToEndId	Max35Text	Optional	Unique identification assigned by the initiating party to unambiguously identify the transaction. This identification is passed on, unchanged, throughout the entire end-to-end chain

This data type is used in [oFA-PFDom].

2.98 Payment Product JSON

Attribute	Type	Condition	Description
pathProductName	Max35Text	Mandatory	path definition of the related JSON formatted product, e.g. "sepa-credit-transfers"
supportedJSONPaymentsAttributes	Array of Max35Text	Mandatory	Supported payment attributes (first level attributes) for the addressed product. In [oFA-PFDom] first level attributes for single payments e.g. "paymentIdentification", "creditorAccount",

Attribute	Type	Condition	Description
			"instructedAmount" are defined. These attributes may come with a full data type, e.g. "Payment Identification" or with a restricted data type, e.g. "Payment Identification 1". This array defines, which of these attributes are accepted within this service. It further supports that the related attributes are presented with the potential restriction, e.g. by "paymentIdentification" for the data type "Payment Identification" or by "paymentIdentification_1" for the data Type "Payment Identification 1". Remark: By this definition, the discovery services define implicitly level one and level two of the data structure. It is the choice of the API Clients to ignore the implicit definition of the level 2 definition, if this is not needed for the supported use cases. For bulk or multi-bulk payments data structures defined in the related attributes might be defined on a lower level. However, this field still defines, whether the related attribute is supported (with the related potential restriction in the data type), potentially on a lower level. (IO21)
debtorAccountMandated	Boolean	Mandatory	If true, the debtor account shall be used within this service. if false, the usage of the attribute is optional for the API client.

Attribute	Type	Condition	Description
supportedPurposeCodes	Array of Purpose Code	Mandatory	A list of the supported purpose codes for the related product.
finalStatusCodes	Array of Transaction Status	Mandatory	Final transaction stati for the related product. (IO45)

2.99 Payment Product XML

Attribute	Type	Condition	Description
pathProductName	Max35Text	Mandatory	path definition of the related XML formatted product, e.g. "pain.001-sepa-credit-transfers"
supportedVersions	Array of Max35Text	Mandatory	Additional information on format versions supported, e.g. different XML schemas like "pain.001.01.09"
supportedPurposeCodes	Array of Purpose Code	Mandatory	A list of the supported purpose codes for the related product.
finalStatusCodes	Array of Transaction Status	Mandatory	Final transaction stati for the related product. (IO45)

2.100 Payment Products Text

Attribute	Type	Condition	Description
pathProductName	Max35Text	Mandatory	Path definition of the related text formatted product, e.g. "dtazv" which stands for a legacy German text based payment format for cross-border payments.

Attribute	Type	Condition	Description
			The related values are defined on domestic level only., cp. [oFA-PFDom]
supportedVersions	Max35Text	Optional	The related values are defined on domestic level only., cp. [oFA-PFDom].
finalStatusCodes	Array of Transaction Status	Mandatory	Final transaction stati for the related product. (IO45)

2.101 Person Identification

Attribute	Type	Condition	Description
identification	Max35Text	Mandatory	Unique and unambiguous identification of a oersib.
schemeName Code	Person Identification Code	{Or - Optional	An entry provided by an external ISO code list
schemeName Proprietary	Max35Text	Or – Optional}	A scheme name defined in a proprietary way.
issuer	Max35Text	Optional	Issuer of the identification

2.102 Postal Address

Attribute	Type	Condition	Description
addressLines	Array of Max140Text	{Or - Optional	At most seven entries are permitted. May only be used, if none of the structured address elements "streetName",

Attribute	Type	Condition	Description
			"buildingNumber", "postcode" or "townName" is used. Remark: For SEPA transactions this is further restricted to a maximum of 1 entry.
department	Max70Text	Or – Optional}	Identification of a division of a large organisation or building.
subDepartment	Max70Text		Identification of a sub-division of a large organisation or building.
streetName	Max70Text		Name of a street or thoroughfare.
buildingNumber	Max16Text		Number that identifies the position of a building on a street.
buildingName	Max35Text		Name of the building or house.
floor	Max70Text		Floor or storey within a building.
postBox	Max16Text		Numbered box in a post office, assigned to a person or organisation, where letters are kept until called for.
room	Max70Text		Building room number.
postCode	Max16Text		Identifier consisting of a group of letters and/or numbers that is added to a postal address to assist the sorting of mail.
townName	Max35Text		Name of a built-up area, with defined boundaries, and a local government. Usage Rule: If address lines are not used, this attribute is mandatory.
townLocationName	Max35Text		Specific location name within the town.

Attribute	Type	Condition	Description
districtName	Max35Text		Identifies a subdivision within a country sub-division.
countrySubDivision	Max35Text		Identifies a subdivision of a country such as state, region, county.
country	Country Code	Optional	Nation with its own government. Usage Rule: If address lines are not used, this attribute is mandatory.

Remark: The usage of addressLines will be phased out in November 2025 at least for SEPA payments.

Remark: The elements of the structured address information marked by "Optional Or}" may only be used, if the element "addressLines" is not used.

2.103 Private Identification

Attribute	Type	Condition	Description
birthDate	ISODate	Optional	
provinceOfBirth	Max35Text	Optional	
cityOfBirth	Max35Text	Optional	
countryOfBirth	Country Code	Optional	
others	Array of Person Identification	Conditional	Unique identification of a person, as assigned by an institution, using an identification scheme. Shall be used if none of the above attributes is used.

2.104 Product Rights

Attribute	Type	Condition	Description
product	Payment Product	Mandatory	payment product definitions as used in the URI for payment initiation
signingClass	Singing Class	Mandatory	
limits	Array of Limit	Optional	Limits applying to the authorisation of related payment products.

2.105 Proxy Account Identification

Attribute	Type	Condition	Description
typeCode	Proxy Account Type Code	{Or optional -	Type of the proxy identification
typeProprietary	Max35Text	Or optional} –	
identification	Max2048Text	mandatory	

2.106 PSU Credentials

Attribute	Type	Condition	Description
password	String	Conditional	Contains a password in plaintext.
encryptedPassword	String	Conditional	Is used when a password is encrypted on application level.

This data type is used in [oFA-PFSM].

2.107 PSU Data

Attribute	Type	Condition	Description
psu-id	Max140Text	Mandatory	

Attribute	Type	Condition	Description
psu-id-type	PSU Id Type	Optional	
psuName	Max140Text	Mandatory	
psuAddress	Postal Address	Optional	
privateId	Private Identification	Optional	
psu-corporate-id	Max140Text	Optional	
psu-corporate-id-type	PSU Corporate Id Type	Optional	
SCA-Methods	Array of Authentication Type	Optional	
limits	Array of Limit	Optional	Global limits of the PSU.

2.108 PSU Header Data

Attribute	Type	Condition	Description
psu-idRequired	Boolean	Mandatory	The ASPSP is providing whether the PSU-ID is mandatory for all of the following requests (if supported): • Payment Initiation Request • Consent Request • Payment Cancellation • Signing Basket • Subscription for Push AIS (IO15)
psu-id-typeRequired	Boolean	Mandatory	The PSU-ID-Type is necessary if different types of identifiers are allowed by the ASPSP to login and the ASPSP is either not able or does not want to identify the type from the

Attribute	Type	Condition	Description
			content of the field PSU-ID. Usual examples for a PSU-ID-Types are: • Username • Customer reference number • Account reference, e.g. an IBAN or MSISDN • email address The ASPSP has to decide whether the PSU-ID-TYPE (additional to the PSU-ID) is mandatory in all of the following requests where supported • Payment Initiation Request • Consent Request • Payment Cancellation • Signing Basket • Subscription Request (IO16)
supportedPsu-id-types	Array of PSU ID Type	Conditional	Shall be supported, if psu-id-typeRequired equals true.
psu-corporate-idRequired	Boolean	Mandatory	PSU-Corporate-ID required in transaction initiation message, if a corporate account is affected (IO19)
psu-corporate-id-typeRequired	Boolean	Mandatory	PSU-Corporate-ID-TYPE re-quired in message, if a corporate account is affected (IO20)
supportedPsu-corporate-id-types	Array of PSU Corporate ID Type	Conditional	Shall be supported, if psu-corporate-id-typeRequired equals true.

2.109 PSU Id Type

Attribute	Type	Condition	Description
psu-id-type	Max35Text	Mandatory	A supported PSU-Id-Type.
description	Max250Text	Optional	Additional description of the addressed PSU-Id-Type.

2.110 PSU Corporate Id Type

Attribute	Type	Condition	Description
psu-corporate-id-type	Max35Text	Mandatory	A supported PSU-Corporate-Id-Type.
description	Max250Text	Optional	Additional description of the addressed PSU-Corporate-Id-Type.

2.111 Push Account Entry Parameters

Attribute	Type	Condition	Description
accountEntryCriteria	Push Account Entry Trigger Criteria	Optional	This attribute is defining the account entries to be pushed to the API Client push URI: All entries which are fulfilling all the criteria listed in this attribute will be pushed. If the attribute is not provided, all account entries will be pushed.
acceptedFormat	Mime Type Code	Mandatory	This attribute describes the mime type which is supported for pushing the account entries by the API Client. If there is no match with the mime types offered by the ASPSP, the subscription is rejected with a dedicated error code.
preferredAttributes	Array of Max35Text	Optional	These attributes are the preferred ones for every account entry to be pushed. The ASPSP might ignore

Attribute	Type	Condition	Description
			this preference. Only attributes of the first level of Type Transactions as defined in [oFA-IG-Com] might be used.
documentsPreferred	Boolean	Optional	RFU: This data element indicates that the API Client prefers to also receive documents attached to an account entry. This data attribute might be ignored by the ASPSP.

This data type is used in [oFA PAIS].

2.112 **Push** Account Entry Trigger Criteria

Attribute	Type	Condition	Description
bankTransactionCodePatterns	Array of Bank Transaction Code Pattern	optional	The criterion is fulfilled if the bankTransactionCode of the related account entry matches one of the bank transaction codes or code patterns (see the following paragraph) defined by this list. Any particle of the Bank Transaction Code triple might be replaced by the "???" character, indicating that all sub family codes apply. However, at least one particle of the triple must be stated explicitly (e.g. "???-CCRD-???").
proprietaryBankCodes	Array of Max35Text	Optional	The criterion is fulfilled if the proprietaryBankTransactionCode of the related account entry equals one of the proprietary bank transaction codes defined by this list.
accountEntryStatus	Account Entry Status	Optional	The criterion is fulfilled if the account entry has the related

Attribute	Type	Condition	Description
			status booked or pending respectively. Note: Entries might be pushed twice in case where this criterion is not provided if the transactionStatus is changed from pending to booked in the account system.
maximumAmount	Amount	Optional	The criterion is fulfilled if the (absolute value of the) transactionAmount of the related account entry is provided in the same currency and if the transactionAmount is less or equal to the amount provided in this attribute.
minimumAmount	Amount	Optional	The criterion is fulfilled if the (absolute value of the) transactionAmount of the related account entry is provided in the same currency and if the transactionAmount is greater or equal to the amount provided in this attribute.
debtorAccount	Account Reference	Optional	The criterion is fulfilled if the counterparty account of the related account entry is a debtor account and equals the account defined in this attribute.
creditorAccount	Account Reference	Optional	The criterion is fulfilled if the counterparty account of the related account entry is a creditor account and equals the account defined in this attribute.
creditDebitIndicator	Credit Debit Code	Optional	The criterion is fulfilled if the related account entry is a credit as defined

Attribute	Type	Condition	Description
			in this attribute, or a debit respectively.
endToEndId	Max35Text	Optional	The criterion is fulfilled if the endToEndId of the related accountpayment entry equals the string provided in this attribute.
purpose	Purpose Code	Optional	The criterion is fulfilled if the related account entry has a purpose code equal to the code provided in this attribute.
remittanceInformation Unstructured	Max140Text	Optional	The criterion is fulfilled if the related account entry has an unstructured remittance information where the string provided in this attribute is contained as a substring. When comparing substrings, the usage of small or capital letters is ignored.

2.113 Push Account Statements Parameters

Attribute	Type	Condition	Description
accountStatementCriteria	Push Account Statement Trigger Criteria	Mandatory	This attribute is defining the account statements to be pushed to the API Client push URI: Every time, the criteria listed in this attribute are all fulfilled, the current account statement of the related account is pushed.

This data type is used in [oFA PAIS].

2.114 Push Account Statement Trigger Criteria

Attribute	Type	Condition	Description
reports	Max35Text	{OR	One of the values ""MT942" or one of the camt.052 versions offered in

Attribute	Type	Condition	Description
			the Discovery API should be used, e.g. "camt.052.001.02", "BG.JSON2.0". " The criterion is fulfilled if a related report is available.If the related report format is not supported, then the ASPSP is rejecting the subscription with a dedicated error code.
notifications	Max35Text	OR	One of the values "MT941", "MT900", "MT910" or camt.054 versions. If the related notification format is not supported, then the ASPSP is rejecting the subscription with a dedicated error code.
statements	Max35Text	OR}	One of the values "MT940", and "camt.053", "BG.JSON2.0" shall be used. The criterion is fulfilled if a related statement is available. If the related statement format is not supported, then the ASPSP is rejecting the subscription with a dedicated error code.
event	Max35Text	Optional	Only the value "afterGeneration" may be used. The criterion is fulfilled if the related report is generated and made available to the PSU in the online channel.
times	Array of ISO Time	Optional	The value is provided in format "hh:mm:ss". The criterion is fulfilled if the current local ASPSP time is matching one of the defined times in the list of times. The list shall not be empty.

2.115 Push Balance Parameters

Attribute	Type	Condition	Description
balanceCriteria	Push Balance Trigger Criteria	Mandatory	This attribute is defining the balance information to be pushed to the API Client push URI: Every time, a balance is fulfilling all the criteria listed in this attribute it is pushed.

This data type is used in [oFA PAIS].

2.116 Push Balance Trigger Criteria

Attribute	Type	Condition	Description
balanceAmount	Amount	Optional	The criterion is fulfilled if the current balance of type balanceType provided below fulfills the balanceOperator requirement provided below together with the provided balanceAmount in this attribute and where the last provided balance did not fulfill it.
balanceOperator	Max35Text	Optional	Must be one of the following codes: • less, • lessOrEqual, • greaterOrEqual, • greater
balanceType	Balance Type	Optional	This criterion is fulfilled if the balance is of this type. If the related balance type is not supported for the addressed account, then the ASPSP is rejecting the subscription with a dedicated error code.
creditLimitIncluded	Boolean	Optional	This criterion is fulfilled if the related current balance is provided with the credit limit included (if this

Attribute	Type	Condition	Description
			attribute is true) or not (if this attribute is false). If the attribute is true and if the related balance type defined above is not applicable to the condition whether a credit limit is included then the subscription is rejected with a dedicated error code.
times	Array of ISO Time	Optional	The value is provided in format "hh:mm:ss". The criterion is fulfilled if the current local ASPSP time is matching one of the defined times in the list of times. The list shall not be empty.

2.117 Push RTP Parameters

Attribute	Type	Condition	Description
rtpCriteria	Push RTP Trigger Criteria	Optional	This attribute is defining the incoming RTP requests to be pushed to the API Client push URI: All RTPs which are fulfilling all the criteria listed in this attribute will be pushed. If the attribute is not provided, all RTPs will be pushed as soon as they are arriving at the ASPSP's system.
acceptedFormat	Mime Type Code	Mandatory	This is defining the mime type. If the requested mime type is not supported by the ASPSP, the request is rejected. If not used, then the ASPSP is choosing the mime type.

Attribute	Type	Condition	Description
			At a minimum the ASPSP shall support a JSON mime type.
documentsPreferred	Boolean	Optional	RFU: This data element indicates that the API Client prefers to also receive documents attached to an RTP. This data attribute might be ignored by the ASPSP.

This data type is used in [oFA PAIS].

2.118 **Push** RTP Trigger Criteria

Attribute	Type	Condition	Description
event	Max35Text	Optional	"afterReception" is the only supported value.
timeToExpire	Integer	Optional	only forward incoming RTP with an expiry timestamp within that period, where the period is defined by the number of minutes as provided in this attribute. This criterion is meant to forward only short term RTPs.
expiryReminder	Boolean	Optional	This criterion is fulfilled if a long term RTP is still not confirmed 2 days before the RTP expires.

2.119 Remittance

The remittance type will be re-defined in version 2 to be able to support more complex requirements. This is not downwards compatible.

Attribute	Type	Condition	Description
referredDocumentInformation	Array of Referred	Optional	Set of elements used to identify the documents

Attribute	Type	Condition	Description
	Document Information		referred to in the remittance information.
creditorReferenceInformation	Creditor Reference Information	Optional	Reference information provided by the creditor to allow the identification of the underlying documents.
additionalRemittanceInformation	Array of Max140Text	Optional	Additional information to be displayed to the PSU e.g. within authorisation processes. This will not be used in related interbank payments.

This data type is used in [oFA-PFDom], [oFA-IG-EPIS].

2.120 References

Attribute	Type	Condition	Description
accountServicerReference	Max35Text	Optional	
instructionId	Max35Text	Optional	Unique identification as assigned by an instructing party for an instructed party to unambiguously identify the instruction.
endToEndId	Max35Text	Optional	
uetr	UUID	Optional	Universally unique identifier to provide an end-to-end reference of a payment transaction. It is recommended to use time-based version (version 1) and variant 8,9,a or b of the UUID as defined in [RFC4122] for UUID fields such as uetr.
mandateId	Optional	Optional	direct debit related mandate.

Attribute	Type	Condition	Description
checkNumber	Max35Text	Optional	

2.121 Referred Document Information

Attribute	Type	Condition	Description
typeCode	Referred Document Type Code	{Or - Optional	Specifies the type of referred document, provided as code.
typeProprietary	Max35Text	Or – Optional}	Specifies the type of referred document, provided in proprietary encoding.
typeIssuer	Max35Text	Optional	Issuer of the document type.
number	Max35Text	Optional	Unique and unambiguous identification of the referred document.
relatedDate	ISO Date	Optional	Date associated with the referred document.

2.122 Referred Mandate Document Information

Attribute	Type	Condition	Description
typeCode	Reference Document Type Code	{Or – Optional	Specifies the type of referred document, provided as code.
typeProprietary	Max35Text	Or – Optional}	Specifies the type of referred document, provided in proprietary encoding.
typeIssuer	Max35Text	Optional	Issuer of the document type.
number	Max35Text	Optional	Unique and unambiguous identification of the referred document.
creditorReference	Max35Text	Optional	Unique and unambiguous identification as assigned by the creditor to the referred document shared with the debtor for its own reference.

Attribute	Type	Condition	Description
relatedDate	ISO Date	Optional	Date associated with the referred document.

2.123 Regulatory Reporting

Attribute	Type	Condition	Description
debitCreditReportingIndicator	Regulatory Reporting Type Code	Optional	Identifies whether the regulatory reporting information applies to the debit side, to the credit side or to both debit and credit sides of the transaction.
detailsCode	Max35Text	Optional	Specifies the nature, purpose, and reason for the transaction to be reported for regulatory and statutory requirements in a coded form.
detailsInformation	Array of Max35Text	Optional	Additional details that cater for specific domestic regulatory requirements.

This data type is used in [oFA-PFDom].

2.124 Report Exchange Rate

Attribute	Type	Condition	Description
sourceCurrency	Currency Code	Mandatory	Currency from which an amount is to be converted in a currency conversion.
exchangeRate	String	Mandatory	Factor used to convert an amount from one currency into another. This reflects the price at which one currency was bought with another currency.
unitCurrency	Currency Code	Mandatory	Currency in which the rate of exchange is expressed in a currency exchange. In

Attribute	Type	Condition	Description
			the example 1EUR = xxxCUR, the unit currency is EUR.
targetCurrency	Currency Code	Mandatory	Currency into which an amount is to be converted in a currency conversion.
quotationDate	ISODate	Mandatory	Date at which an exchange rate is quoted.
contractIdentification	Max35Text	Optional	Unique identification to unambiguously identify the foreign exchange contract.

2.125 RTP Details

Attribute	Type	Condition	Description
paymentConditions	Payment Conditions	optional	Conditions for the execution of the payment.
creationDateTime	ISODateTime	optional	creationDateTime of the originator
requestedExpiry DateTime	ISODateTime	mandatory	Date and Time at which this request expires.
originatorPspld	Financial Institution Identification	Optional	Originator payment service provider in case of indirect participation regarding a related API Access Scheme or bilateral contract

2.126 SCA Approach Methods Mapping

Attribute	Type	Condition	Description
scaApproach	SCA Approach Type	Mandatory	
scaMethods	Array of Authentication Type	Mandatory	

2.127 Securities Account Fee

Represents a Fee rule that is applicable to a securities account:

Attribute	Type	Condition	Description
typeCode	Securities Fee Type Code	{Or	Type of the fee as a code. Might be adjusted in the future based on market experience.
typeProprietary	Max35Text	Or}	Type of the fee as a proprietary string.
feeRules	Array of Securities Account Fee Rule	Mandatory	Array of fee rules. Each rule represents how the amount of the fee is determined either explicitly by stating the amount or implicitly by providing a percentage, minimum and maximum amount. This array will usually only contain one entry to represent the whole rule. Only in the case of the fee being calculated as a tiered percentage, several entries are included to represent the several tiers of the fee rule.
applicableFrom	ISODate	Optional	First date, when this fee is applicable. If not present, the fee is applicable starting with the creation of this account / the latest reporting date supported by the ASPSP.
applicableTo	ISODate	Optional	Last date, when this fee is applicable.

Attribute	Type	Condition	Description
			If not present, the fee is applicable indefinitely.
additionalInformation	Max500Text	Optional	

2.128 Securities Account Fee Rule

Attribute	Type	Condition	Description
amount	Amount	{Or	Amount of the fee.
percentage	String	Or}	Percentage of the fee. Up to 20 significant figures. The decimal separator is a dot. Trailing zeroes must be truncated. Fee as a percentage.
fromBaseAmount	Amount	Optional	To be used for fees where a tiered percentage is used to calculate the fee relative to an implicitly defined base amount (such as a transaction's volume being the base amount for a courtage). The range of one tier is defined as all applicable entities with base amounts X and $\text{fromBaseAmount} \leq X \leq \text{toBaseAmount}.$ In any array of fee rules, only one of these entries may be missing the "fromBaseAmount" element
toBaseAmount	Amount	Optional	

Attribute	Type	Condition	Description
			(indicating that the rule represented by this entry is used without a lower bound to the base amount) and only one entry may be missing the "toBaseAmount" element (indicating that the rule represented by this entry is applied without an upper limit to the base amount). In cases, where the fee is not calculated as a tiered percentage, only one entry in array "feeRules" is included. This entry must be missing both subelements "fromBaseAmount" and "toBaseAmount".
minimumAmount	Amount	Optional	Minimum amount of the fee in each billing period / for each billed event if applicable.
maximumAmount	Amount	Optional	Maximum amount of the fee in each billing / for each billed event period if applicable.

2.129 Securities Order

Attribute	Type	Condition	Description
orderId	String	Optional	Can be used as access-ID in the API, where more details on an order is offered. If this data attribute is provided, this shows that the API Client can get access on more details about this transaction using the Read

Attribute	Type	Condition	Description
			Securities Order Details Request. Remark: Implementers might want to anticipate future services supporting placement or revocation of orders via the openFinance API. It is therefore recommended to define the orderId in a way that it will also identify an order in the case of a placement / revocation.
side	Securities Order Side	Mandatory	Type of transaction that is associated with this order.
financialInstrument	Financial Instrument	Mandatory	Financial instrument the order refers to.
unitsNumber Order	Number	{Or	Nominal or numeric quantification of the financial instrument that is ordered.
unitsNominal Order	Amount	Or}	
unitsNumber Display	Number	{Or – Optional	Nominal or numeric quantification of the order that shall be displayed.
unitsNominal Display	Amount	Or - Optional}	
placeOfTrade	Market Identification	Optional	
limitPrice Amount	Amount	{Or- Optional	Limit price displayed as an amount.
limitPrice Percent	String	Or – Optional}	Limit price displayed as a percentage rate. Up to 20 significant figures. The decimal separator is a dot.

Attribute	Type	Condition	Description
			Trailing zeroes must be truncated.
stopPrice Amount	Amount	{Or- Optional	Stop price displayed as an amount.
stopPrice Percent	String	Or – Optional}	Stop price displayed as a percentage rate. Up to 20 significant figures. The decimal separator is a dot. Trailing zeroes must be truncated.
tradingSessionIndicator	Trading Session Type Code	Optional	
typesOfOrder	Array of Type of Order Code	Optional	One or more codes to indicate characteristics of the order.
timeInForce	Order Time Limit Code	Optional	If the order is restricted by a time limit, the type time limit is indicated by a value of the corresponding code set.
expiryDate	ISODate	{Or - Optional	Expiry date of the order if applicable.
expiryDateTime	ISODateTime	Or – Optional}	Expiry date and time of the order if applicable.
relatedCashAccount	Account Reference	Optional	If the execution of this order implies the transfer of money, cash account to be affected by this transfer from the PSU's side.
orderSplit	Boolean	Optional	Value true indicates that this order is the result of an order split. Missing values indicate "false".

Attribute	Type	Condition	Description
orderModifiable	Boolean	Optional	Value true indicates that this order can (still) be modified. Missing values indicate "false".
orderStatus	Order Status Code	Mandatory	
details	Max500Text	Optional	Additional details to the position
_links	Links	Optional	The following types of links are supported: "relatedOrders": An array of links to request order details of related orders (e.g. other orders that originated from the same order split as this order). "relatedTransactions": An array of links to request securities transaction details of transactions that resulted from this order. "orderDetails": Single link for retrieving details on this specific order.

2.130 Securities Position

This data structure represents one position within a securities account:

Attribute	Type	Condition	Description
financialInstrument	Financial Instrument	Mandatory	Financial Instrument that is contained in this position.
unitsNumber	Number	{Or	Nominal or numeric quantification of the financial instrument within this position,
unitsNominal	Amount	Or}	

Attribute	Type	Condition	Description
			negative values for short positions.
externalIdentifier	Max35Text	Optional	Name or identifier with an intrinsic meaning for the PSU to be displayed.
safekeepingPlace	BICFI	Optional	BIC of the place where the securities are safe-kept, physically or notionally.
safekeepingCountry	Country Code	Optional	Country where the securities are safe-kept.
balanceType	SecuritiesBalanceTypeCode	Optional	Specifies the nature of the securities or investment fund balance. Type of balance in case of multiple positions for the same financial instrument for further distinction. Remark: In consequence, several positions with the same financial instrument may occur in one position list, having different values of balanceType".
averageBuyingPrice	Amount	Optional	Average buying price of the position excluding any fees or taxes.
averageSellingPrice	Amount	Optional	Average selling price of the position, e.g. in case of a short position Price excluding any fees or taxes.
totalBuyingPrice	Amount	Optional	In case of bonds, an average buying price does not make sense. Therefore, the total buying price excluding any

Attribute	Type	Condition	Description
			fees or taxes can be displayed instead.
estimatedCurrentValue	Evaluated Amount	Optional	Estimated value of the position and timestamp of the estimation.
accruedInterest	Accrued Interest	Optional	Accrued interest by the position.
currencyExchange	Array of Report Exchange Rate	Optional	The ASPSP might include exchange rates e.g. if the security is denoted in another currency than the account itself.
details	Max500Text	Optional	Additional details describing the position.

2.131 Securities related Date or Time

Attribute	Type	Condition	Description
type	String	Mandatory	Type of the date / date time. The following values are pre-defined: • effectiveSettlementDate, • settlementDate, • valueDate, • performanceDate, • bookingDate, • transactionDate
date	ISODate	{Or	
dateAndTime	ISODateTime	Or}	

2.132 Securities Transaction

Attribute	Type	Condition	Description
transactionId	String	Optional	Can be used as access-ID in the API, where more details on a transaction is offered. If this data attribute is provided, this shows that the API Client can get access on more details about this transaction using the Read Transaction Details Request.
entryReference	Max35Text	Optional	Identification of the transaction as used e.g. for reference for deltafunction on application level.
relevantDates	Array of Securities related Date or Time	Mandatory	At least one entry containing a pre-defined date type must be present.
financialInstrument	Financial Instrument	Mandatory	Financial instrument that was transferred within the transaction.
orderId	String	Optional	Resource Id of the order resource that triggered this transaction, if applicable.
unitsNumber	Number	{Or	Nominal or numeric quantification of the financial instrument that has been transferred within this transaction. Negative values indicate that the respective quantity of the financial instrument has been taken from the securities account, positive values indicate that the quantity has been added.
unitsNominal	Amount	Or}	

Attribute	Type	Condition	Description
transactionType Code	Transaction Activity Code	{Or	Type of the transaction as code or as a proprietary string.
transactionType Proprietary	Max35Text	Or}	
placeOfTrade	Market Identification	Optional	
amountIncludes Fees	Boolean	Optional	Indicates whether the transactionAmount (see below) is including fees. Default: false.
amountIncludes Taxes	Boolean	Optional	Indicates whether the transactionAmount (see below) is including taxes. Default: false.
transactionAmount	Amount	Optional	Transaction amount for the transferred security for the PSU. Potential fees or taxes are reflected as indicated by elements "amountIncludesFees" and "amountIncludesTaxes". Remark: The price denoted here is the price for the whole lot of securities within this transaction, not for a single item.
relatedFees	Array of Securities Transaction Fee	Optional	Only allowed if amountIncludesFees = true. In that case, the structure contains details of the fees that have been applied to this transaction (and therefore represent additional costs of the transaction beyond the transactionAmount).

Attribute	Type	Condition	Description
			Fees with positive amount are debited by the PSU, only in the rare case of a fee credited to the PSU (e.g. in case of a reversal) negative amounts are used.
currencyExchange	Array of Report Exchange Rate	Optional	The ASPSP might include exchange rates e.g. if the transaction has been settled in another currency than the PSU's currency.
reversalIndicator	Boolean	Optional	Indicates whether it is the reversal of a previously reported movement. Default: false
reversedTransactionId	String	Optional	"transactionId" of the reversed transaction, if applicable and supported by the ASPSP.
unitsNumberBeforeTx	Number	{Or - Optional	Nominal or numeric quantification of the financial instrument within the primary position before the transaction, negative values for short positions.
unitsNominalBeforeTx	Amount	Or - Optional }	
unitsNumberAfterTx	Number	{Or - Optional	Nominal or numeric quantification of the financial instrument within the primary position after the transaction, negative values for short positions.
unitsNominalAfterTx	Amount	Or - Optional }	
accruedInterest	Accrued Interest	Optional	Accrued interest of the position to make transparent what the PSU really lost within this transaction.
details	Max500Text	Optional	Additional details to the transaction.

Attribute	Type	Condition	Description
_links	Links	Optional	The following types of links are supported: "transactionDetails": Single link for retrieving details on this specific transaction.

2.133 Securities Transaction Fee

Attribute	Type	Condition	Description
typeCode	Securities Fee Type Code	{Or	
typeProprietary	Max35Text	Or}	
amount	Amount	Mandatory	
_links	Links	Optional	Links to resources, where the ASPSP can provide additional information on the fee, e.g. an overview on the general fee rules of the underlying securities account. Currently no specific href types are defined in this context.

2.134 Source of Price

Attribute	Type	Condition	Description
type	Market Type Code	Mandatory	
mic	Max4Text	{Or - Optional	If the type indicates local Market (Code: "LMAR"), the identification of the market should additionally be included here.
market Identifier Proprietary	Max35Text	Or - Optional }	

Attribute	Type	Condition	Description
			Technically, Source of Price is an extension of Market Identification.

2.135 Standing Order Details

Attribute	Type	Condition	Description
startDate	ISODate	Mandatory	The first applicable day of execution starting from this date the first payment was/will be executed.
endDate	ISODate	Optional	The last applicable day of execution If not given, it is an infinite standing order.
executionRule	String	Optional	"following" or "preceding" supported as values. This data attribute defines the behavior when a transaction date resulting from a standing order falls on a weekend or bank holiday. The payment is then executed either the "preceding" or "following" working day.
withinAMonthFlag	Boolean	Optional	This element is only used in case of frequency equals "Monthly". If this element equals false it has no effect. If this element equals true, then the execution rule is overruled if the day of execution would fall into a different month using the execution rule. Example: executionRule equals "preceding", dayOfExecution equals "02" and the second of a month is a Sunday. In this case, the transaction date would be on the last day of the month before. This would be overruled if withinAMonthFlag equals true and the payment is processed on Monday the third of the Month.

Attribute	Type	Condition	Description
			Remark: This attribute is rarely supported in the market.
frequency	Frequency Code	Mandatory	The frequency of the recurring payment resulting from this standing order.
monthsOfExecution	Array of Max2Text	Conditional	The format is following the regular expression <code>\d{1,2}</code>. The array is restricted to 11 entries. The values contained in the array entries shall all be different and the maximum value of one entry is 12. This attribute is contained if and only if the frequency equals "MonthlyVariable". Example: An execution on January, April and October each year is addressed by ["1", "4", "10"].
multiplicator	Numerical	Optional	This is multiplying the given frequency resulting the exact frequency, e.g. Frequency=weekly and multiplicator=3 means every 3 weeks. Remark: This attribute is rarely supported in the market.
dayOfExecution	Max2Text	Optional	"31" is ultimo. The format is following the regular expression <code>\d{1,2}</code>. Example: The first day is addressed by "1". The date is referring to the time zone of the ASPSP.
limitAmount	Amount	Conditional	limitAmount Amount limit for fund skimming, e.g. skim all funds above this limit to savings account, i.e. typically a specific periodic payments with fixed remaining amount rather than

Attribute	Type	Condition	Description
			fixed transaction amount. Amount may be zero as well as below zero, i.e. negative. Constraints: transactionAmount needs to be zero and bankTransactionCode needs to specify PMNT-MCOP-OTHR for fund skimming

2.136 Structured Additional Transaction Information

Attribute	Type	Condition	Description
transactionStatus	Transaction Status	Mandatory	
transactionCreated	User Entry	Optional	
transaction Authorised	Array of User Entry	Optional	
transaction Cancelled	Array of User Entry	Optional	

2.137 Structured Additional Information

Attribute	Type	Condition	Description
standingOrderDetails	Standing Order Details	{OR	Details of underlying standing orders.
rtpDetails	RTP Details	OR}	Details of underlying incoming RTPs

2.138 Subscription Entry

Attribute	Type	Condition	Description
accountId	Account Reference	Mandatory	This is the account from which related entries shall be pushed.

Attribute	Type	Condition	Description
validUntil	ISODate	Optional	Requested validity period of the service.
subscriptionEntryName	Max35Text	Optional	Name of the subscription entry
subscriptionEntryId	String	Conditional	Forbidden in HTTP Requests, e.g. <i>Initiate Subscription Request</i> and in an <i>Add a Subscription Entry Request</i> by the TPP (as this is assigned by the ASPSP). Mandatory in each entry of a response from an ASPSP.
apiClientPrimaryPush URI	Max256Text	Mandatory	The URI where related account information shall be pushed to.
apiClientSecondaryPush URI	Max256Text	[Optional if supported]	The URI where related account information shall be pushed to if the primary push URI is not reachable. If this attribute is used by the API Client and the ASPSP is not supporting this feature, then the request will be rejected with a dedicated additional error information.
callbackWithLinkPreferred	Boolean	Optional	API Client prefers to receive hyperlinks pointing to the related account information element if the related subservice criteria are met.
callbackWithStaticTextPreferred	Boolean	Optional	API Client prefers to get informed by static text if the related subservice criteria are met.
staticCallbackText	Max140Text	Optional	The text to be provided by the ASPSP in case static text is

Attribute	Type	Condition	Description
			sent if the related subservice criteria are met.
pushAccountEntryParameters	Push Account Entry Parameters	{Or	Parameters for a subscription entry for the Push Account Entry Service.
pushAccountStatementParameters	Push Account Statement Parameters	Or	Parameters for a subscription entry for the Push Account Statement Service.
pushBalanceParameters	Push Balance Parameters	Or	Parameters for a subscription entry for the Push Balance Information Service.
pushRtpParameters	Push RTP Parameters	Or}	Parameters for a subscription entry for the Push RTP Information Service.

This data type is used in [oFA PAIS].

2.139 Subscription Identification and Link

Attribute	Type	Condition	Description
subscriptionId	Max70Text	Mandatory	Identification of the generated subscription.
subscriptionStatus	Max35Text	Mandatory	Status of the related subscription.
_links	Links	Mandatory	A hyperlink of type "subscription".

This data type is used in [oFA PAIS].

2.140 Transactions

Attribute	Type	Condition	Description
transactionId	String	Optional	Can be used as access-ID in the API, where more details on an transaction is offered. If this data attribute is provided this shows that the AIS can get access on more details about this transaction using the GET Transaction Details Request. Remark: ASPSPs should ensure that transactionID, if provided, is unique relative to the account that PSU is accessing, in order to ensure the quality of the data provided to the API Client.
entryReference	Max35Text	Optional	Is the identification of the transaction as used e.g. for reference for delta function on application level. The same identification as for example used within camt.05x messages.
batchIndicator	Boolean	Optional	If this indicator equals <code>true</code> , then the related entry is a batch entry.
batchNumberOf Transactions	Integer	Conditional	Shall be used if and only if the <code>batchIndicator</code> is contained and equals <code>true</code> .
references	References	Optional	Might contain the payment identification attributes <code>endToEndId</code> as well as the new UETR field.

Attribute	Type	Condition	Description
			In addition it is including accountServicerReference, mandateId, checkId (all level 1 fields in V1.3.x).
localInstrumentCode	Local Instrument Code	{Or Optional}	User community specific instrument as a code provided by ISO20022.
localInstrumentProprietary	Max35Text	Or Optional}	User community specific instrument as a proprietary attribute.
bookingDate	ISODate	Optional	The Date when an entry is posted to an account on the ASPSPs books.
valueDate	ISODate	Optional	Date at which assets become available to the account owner in case of a credit entry, or cease to be available to the account owner in case of a debit entry. Usage: If entry status is pending and value date is present, then the value date refers to an expected/requested value date.
transactionAmount	Amount	Mandatory	The amount of the transaction or batch as billed to the account.
currencyExchange	Array of Report Exchange Rate	Optional	
amountDetails	Amount Details	Optional	Additional information to the transaction amount.

Attribute	Type	Condition	Description
interbankSettlement Date	ISODate	Optional	Date on which the amount of money ceases to be available to the agent that owes it and when the amount of money becomes available to the agent to which it is due.
cardTransaction	Card Transaction	Optional	Card transaction details in case of an underlying card transaction.
creditor	Party Description	Optional	Name and potentially an identification of the creditor if a "Debited" transaction
creditor Account	Account Reference	Conditional	Account of the payee.
creditorAgent	Agent Description1	Optional	Proprietary identifications might be provided under financialInstitutionId/other
ultimate Creditor	Party Description4	Optional	Only name and identification is supported for now for this party.
debtor	Party Description	Optional	Name and potentially an identification of the debtor if a "Credited" transaction
debtorAccount	Account Reference	Conditional	
debtorAgent	Agent Description1	Optional	Proprietary identifications might be provided under financialInstitutionId/other
ultimateDebtor	Party Description4	Optional	
remittance Information Unstructured	Array of Max140Text	Optional	

Attribute	Type	Condition	Description
remittance Information Structured	Array of Remittance	Optional	
entryDetails	Array of Entry Details	Optional	Might be used by the ASPSP to transport details about transactions within a batch.
additionalTransactionInformation	Max500Text	Optional	Might be used by the ASPSP to transport additional transaction related information to the PSU
additionalInformation Structured	Structured Additional Information	Conditional	Is used if and only if the bookingStatus entry equals "information". Every active standing order related to the dedicated payment account result into one entry. Each received RTP result into one entry.
additionalTransaction informationStructured	Structured Additional Transaction Information	Optional	Data about the transactional process of submission, authorisation and/or cancellation of the underlying payment.
purposeCode	Purpose Code	Optional	
bank TransactionCode	Bank Transaction Code	Optional	Bank transaction code as used by the ASPSP and using the sub elements of this structured code defined by ISO20022.

Attribute	Type	Condition	Description
			For standing order reports the following codes are applicable: "PMNT-ICDT-STD0" for credit transfers, "PMNT-IRCT-STD0" for instant credit transfers "PMNT-ICDT-XBST" for cross-border credit transfers "PMNT-IRCT-XBST" for cross-border real time credit transfers and "PMNT-MCOP-OTHR" for specific standing orders which have a dynamical amount to move left funds e.g. on month end to a saving account
bank TransactionCodeProprietary	Max35Text	Optional	proprietary bank transaction code as used within a community or within an ASPSP e.g. for MT94x based transaction reports. Renaming of the attribute to adapt to statements.
balanceAfter Transaction	Balance	Optional	This is the balance after this transaction. Recommended balance type is interimBooked.
_links	Links	Optional	The following links could be used here:

Attribute	Type	Condition	Description
			transactionDetails for retrieving details of a transaction.

This data type is used in [oFA-IG-Com].

2.141 User Parameter

Attribute	Type	Condition	Description
userData	PSU Data	Mandatory	
accountAccessRights	Array of Account Access and Authorisation Rights	Mandatory	

This data type is used in [oFA-Disc].

2.142 User Entry

Attribute	Type	Condition	Description
user	Max70Text	Mandatory	
dateTime	ISODateTime	Mandatory	

3 Code Lists

NOTE: It is indicated below each code, in which openFinance API Framework Implementation Guidelines the related data type is used in attributes of API requests and API responses on **first** level.

3.1 Access Rights Codes

The current list of access right codes. This might be further extended.

Codes	Description
ais	covers details, balances and transactions
accountDetails	covers generic account product information details
balances	Allow to retrieve balances of the addressed account.
transactions	Allow to retrieve transactions of the addressed account.
orders	Allow to retrieve orders from an addressed order book related to a securities account.
ownerName	Allow to retrieve the owner name/owner names related to this account.
psuName	Allow to retrieve the PSU name related to the consent authorisation.
psuLeanIdentification	Allow to retrieve the PSU Name as well as PSU identification related data from the PSU parameters, e.g. birth date, address or passport ID.
trustedBeneficiaries	Allow to retrieve the list of trusted beneficiaries related to this account.
initiatePayments	Allow to initiate payments with the related token.
fundsConfirmations	Allows to access an addressed account for a funds confirmation check.
userParameters	Allows to access the PSU's user parameters applicable to the account configuration.
ibanChecks	Allows to access a restricted version of the user parameters related to signing rights.

Codes	Description
corporateParameters	RFU Will be supported in future.
accountCheckParameters	Allows to access restricted user parameters e.g. role definitions of the PSU related to the account.

3.2 Account Entry Status

Code	Description
booked	Entry is booked on the account.
pending	Entry is pending, funds might be reserved or not.

3.3 Account Reference Code

Attribute	Description
iban	
bban	
msisdn	
pan	
maskedPan	
proxy-<proxytype>	with proxytype being of Type Account Identification Code

3.4 Account Status Code

Code	Description
enabled	account is available
deleted	account is terminated
blocked	account is blocked e.g. for legal reasons

3.5 Balance Type

The following balance types are excluding credit limits unless the `creditLimitIncluded` element is present and equals true in the corresponding balance element.

Remark: This definition is following ISO20022 logic for defining balance types.

Type	Description
closingBooked	Balance of the account at the end of the pre-agreed account reporting period. It is the sum of the opening booked balance at the beginning of the period and all entries booked to the account during the pre-agreed account reporting period. For card-accounts, this is composed of • invoiced, but not yet paid entries
expected	Balance composed of booked entries and pending items known at the time of calculation, which projects the end of day balance if everything is booked on the account and no other entry is posted. For card accounts, this is composed of • invoiced, but not yet paid entries, • not yet invoiced but already booked entries and • pending items (not yet booked)
openingBooked	Book balance of the account at the beginning of the account reporting period. It always equals the closing book balance from the previous report.
interimAvailable	Available balance calculated in the course of the account 'servicer's business day, at the time specified, and subject to further changes during the business day. The interim balance is calculated on the basis of booked credit and debit items during the calculation time/period specified. For card-accounts, this is composed of • invoiced, but not yet paid entries, • not yet invoiced but already booked entries
interimBooked	Balance calculated in the course of the account servicer's business day, at the time specified, and subject to further changes during the business

Type	Description
	day. The interim balance is calculated on the basis of booked credit and debit items during the calculation time/period specified.
forwardAvailable	Forward available balance of money that is at the disposal of the account owner on the date specified.
nonInvoiced	Only for card accounts, to be defined yet.

3.6 Card Data Entry Mode

Type	Description
CICC	EMV based
ECTL	EMV based contactless
MGST	Magnetic Stripe
PHYS	Manual Entry
BRCD	Barcode

This is following an ISO20022 Code: CardsDataReading1Code

3.7 Charge Bearer

Type	Description
DEBT	All transaction charges are to be borne by the debtor.
CRED	All transaction charges are to be borne by the creditor.
SHAR	In a credit transfer context, means that transaction charges on the sender side are to be borne by the debtor, transaction charges on the receiver side are to be borne by the creditor. In a direct debit context, means that transaction charges on the sender side are to be borne by the creditor, transaction charges on the receiver side are to be borne by the debtor.
SLEV	Charges are to be applied following the rules agreed in the service level and/or scheme.

This is following ChargeBearerType1Code from ISO20022.

This code is used in [oFA-PFDom].

3.8 Classification Code

MandateClassification1Code from ISO20022.

3.9 Consent Status

Code	Description
received	The consent data have been received and are technically correct. The data is not authorised yet.
rejected	The consent data have been rejected e.g. since no successful authorisation has taken place.
partiallyAuthorised	The consent is due to a multi-level authorisation, some but not all mandated authorisations have been performed yet.
valid	The consent is accepted and valid for GET account data calls and others as specified in the consent object.
revokedByPsu	The consent has been revoked by the PSU towards the ASPSP.
expired	The consent expired.
terminatedByTpp	The corresponding TPP has terminated the consent by applying the DELETE method to the consent resource.
replacedByTpp	The corresponding TPP has terminated the (recurring) consent implicitly by submitting a new (recurring) consent.

The ASPSP might add further codes. These codes then shall be contained in the ASPSP's documentation of the XS2A interface.

This code is used in [oFA-Co], [oFA-PFSM].

3.10 Consent Type Code

Code	Description
global	This consent type asks certain rights for dedicated account types without further detailing the accounts.

Code	Description
detailed	This consent type asks for access for dedicated accounts and dedicated rights.
aspspManaged	This consent type asks for access on certain account types for dedicated access rights. The accounts provisioned will be added during the authorisation process by the ASPSP.
accountList	This consent type asks for the list of accounts for each addressed account type. In difference to other account information consents, this consent might be authorised by using one authentication factor only, if supported by the ASPSP. In addition to the "accountDetails" access rights, also other non-transactional access rights might be asked for. This might have an impact on the decision of the ASPSP to apply SCA or only one authentication factor of the PSU.

This code is used by [oFA-Co].

3.11 Contact Type

Code	Description
technical	To be contacted in case of items related to the API implementation,
operational	To be contacted in case of topics related to operational issues.
business	To be contacted in case of business related topics on premium services.
security	To be contacted in case of IT security related topics.

3.12 Credit Debit Code

Code	Description
CRDT	Operation is an increase.
DBIT	Operation is a decrease.

3.13 Credit Transfer Payment Method Code

This code set corresponds to ISO 20022 PaymentMethod3Code:

Type	Description
TRF	Credit Transfer
CHK	Cheque
TRA	Transfer Advice Remark: Currently not supported by the openFinance payment data model. The code "TRA" is only mentioned for completeness as it is defined in ISO20022.

This data type is used in [oFA-PFDom].

3.14 Creditor Reference Type Code

This code set is equivalent to the ISO 20022 definition DocumentType3Code:

Type	Description
RADM	Document is a remittance advice sent separately from the current transaction.
RPIN	Document is a linked payment instruction to which the current payment instruction is related, for example, in a cover scenario.
FXDR	Document is a pre-agreed or pre-arranged foreign exchange transaction to which the payment transaction refers.
DISP	Document is a dispatch advice.
PUOR	Document is a purchase order.
SCOR	Document is a structured communication reference provided by the creditor to identify the referred transaction.

3.15 Exchange Rate Type Code

This code set is equivalent to the ISO 20022 definition ExchangeRateType1Code:

Type	Description
SPOT	Exchange rate applied is the spot rate.
SALE	Exchange rate applied is the market rate at the time of the sale.
AGRD	Exchange rate applied is the rate agreed between the parties.

3.16 Frequency Code

The following codes from the "EventFrequency7Code" of ISO 20022 are supported:

- Daily
- Weekly
- EveryTwoWeeks
- Monthly
- EveryTwoMonths
- Quarterly
- SemiAnnual
- Annual
- MonthlyVariable

This code is used in [oFA-IG-Com], [oFA-IG-EPIS].

3.17 Interest Type Code

The following ISO definitions apply for these codes (based on ISO InterestType3code as defined for message type reda.006.001.01, but here, only a restricted code list is supported):

Code	Description
FIXD	ISO 20022: Indicates that the type of interest is fixed. Remark: This is a fixed interest for the period defined either in the interest entry or implicitly by the contract period.
INDE	ISO 20022: Indicates that the type of interest is index. Remark: This rate could relate e.g. to interest rates of central banks.

3.18 Limit Type

Attribute	Description
singleOrder	The limit applies to every transaction.
daily	It is the limit of transactions to be authorised aggregated during a day.
weekly	It is the limit of transactions to be authorised aggregated during a calendar week.
monthly	It is the limit of transactions to be authorised aggregated during a calendar month.

3.19 Mandate Setup Reason Code

This code is covered by the ISO20022 ExternalMandateSetupReason1Code.

NOTE: Currently, there is no external mandate setup reason code defined yet in ISO20022.

3.20 Message Code

The permitted message error codes and related HTTP response codes are listed below. New services may introduce new Message Codes which are not yet reflected here.

3.20.1 Service Unspecific HTTP Error Codes

Message Code	HTTP Response Code	Description
CERTIFICATE_INVALID	401	The contents of the signature/corporate seal certificate are not matching PSD2 general PSD2 or attribute requirements.
ROLE_INVALID	401	The TPP does not have the correct PSD2 role to access this service.
CERTIFICATE_EXPIRED	401	Signature/corporate seal certificate is expired. Remark: If a resource has been created already, a related status reason code is used instead.

Message Code	HTTP Response Code	Description
CERTIFICATE_BLOCKED	401	Signature/corporate seal certificate has been blocked by the ASPSP or the related NCA. Remark: If a resource has been created already, a related status reason code is used instead.
CERTIFICATE_REVOKED	401	Signature/corporate seal certificate has been revoked by QSTP. Remark: If a resource has been created already, a related status reason code is used instead.
CERTIFICATE_MISSING	401	Signature/corporate seal certificate was not available in the request but is mandated for the corresponding. Remark: If a resource has been created already, a related status reason code is used instead.
CLIENT_INVALID	401	Client certificate is valid but the related client is not contained in the related ASPSP or Scheme directory
CLIENT_INCONSISTENT	401	Client information is not consistent with related directory entries
SIGNATURE_INVALID	401	Application layer eIDAS Signature for API Client authentication is not correct.
SIGNATURE_MISSING	401	Application layer eIDAS Signature for API authentication is mandated by the ASPSP but is missing.
ROLE_INVALID	401	The TPP does not have the correct PSD2 role to access this service
FORMAT_ERROR	400	Format of certain request fields are not matching the XS2A requirements. An explicit

Message Code	HTTP Response Code	Description
		path to the corresponding field might be added in the return message. This applies to headers and body entries. It also applies in cases where these entries are referring to erroneous or not existing data instances, e.g. a malformed IBAN.
PARAMETER_NOT_CONSISTENT	400	Parameters submitted by API Client are not consistent. This applies only for query parameters.
PARAMETER_NOT_SUPPORTED	400	The parameter is not supported by the API provider. This code should only be used for parameters that are described as "optional if supported by API provider."
PSU_CREDENTIALS_INVALID	401	The PSU-ID cannot be matched by the addressed ASPSP or is blocked, or a password resp. OTP was not correct. Additional information might be added.
SERVICE_INVALID	400 (if payload) 405 (if HTTP method)	The addressed service is not valid for the addressed resources or the submitted data.
SERVICE_BLOCKED	403	This service is not reachable for the addressed PSU due to a channel independent blocking by the ASPSP. Additional information might be given by the ASPSP.
CORPORATE_ID_INVALID	401	The PSU-Corporate-ID cannot be matched by the addressed ASPSP.
CONSENT_UNKNOWN	403 (if path) 400 (if header)	The Consent-ID cannot be matched by the ASPSP relative to the API Client.
CONSENT_INVALID	401	The consent was created by this TPP but is not valid for the addressed service/resource.

Message Code	HTTP Response Code	Description
CONSENT_EXPIRED	401	The consent was created by this TPP but has expired and needs to be renewed.
TOKEN_UNKNOWN	401	The OAuth2 token cannot be matched by the ASPSP relative to the TPP.
TOKEN_INVALID	401	The OAuth2 token is associated to the TPP but is not valid for the addressed service/resource.
TOKEN_EXPIRED	401	The OAuth2 token is associated to the TPP but has expired and needs to be renewed.
RESOURCE_UNKNOWN	404 (if account-id in path) 403 (if other resource in path) 400 (if payload)	The addressed resource is unknown relative to the API Client. An example for a payload reference is creating a signing basket with an unknown resource identification.
RESOURCE_EXPIRED	403 (if path) 400 (if payload)	The addressed resource is associated with the TPP but has expired, not addressable anymore.
RESOURCE_BLOCKED	400	The addressed resource is not addressable by this request, since it is blocked e.g. by a grouping in a signing basket.
TIMESTAMP_INVALID	400	Timestamp not in accepted time period.
PERIOD_INVALID	400	Requested time period out of bound.
SCA_METHOD_UNKNOWN	400	Addressed SCA method in the Authentication Method Select Request is unknown or cannot be matched by the ASPSP with the PSU.

Message Code	HTTP Response Code	Description
SCA_INVALID	400	Method Application on authorisation resource (e.g. Confirmation Request) blocked since SCA status of the resource equals "failed".
STATUS_INVALID	409	The addressed resource does not allow additional authorisation.

3.20.2 PIS Specific HTTP Error Codes

Message Code	HTTP Response Code	Description
PRODUCT_INVALID	403	The addressed payment product is not available for the PSU .
PRODUCT_UNKNOWN	404	The addressed payment product is not supported by the ASPSP.
PAYMENT_FAILED	400	The payment initiation POST request failed during the initial process. Additional information may be provided by the ASPSP.
KID_MISSING	401	The payment initiation has failed due to a missing KID. This is a specific message code for the Norwegian market, where ASPSP can require the payer to transmit the KID.
EXECUTION_DATE_INVALID	400	The requested execution date is not a valid execution date for the ASPSP.
CANCELLATION_INVALID	405	The addressed payment is not cancellable e.g. due to cut off time passed or legal constraints.

3.20.3 AIS Specific HTTP Error Codes

Message Code	HTTP Response Code	Description
CONSENT_INVALID	401	The consent definition is not complete or invalid. In case of being not complete, the bank is not supporting a completion of the consent towards the PSU. Additional information will be provided.
CONSENT_TYPE_NOT_SUPPORTED	400	The consentType chosen in the consent object is not supported by the ASPSP.
SESSIONS_NOT_SUPPORTED	400	The combined service flag may not be used with this ASPSP.
CONTENT_TEMPORARILY_NOT_AVAILABLE	404	The contents of the response can (temporarily) not be obtained because a service, the interface depends on is currently not available.
ACCESS_EXCEEDED	429	The access on the account has been exceeding the consented multiplicity without PSU involvement per day.
REQUESTED_FORMATS_INVALID	406	The requested formats in the Accept header entry are not matching the formats offered by the ASPSP.

3.20.4 PIIS Specific Error Codes

Message Code	HTTP Response Code	Description
CARD_INVALID	400	Addressed card number is unknown to the ASPSP or not associated to the PSU.
NO_PIIS_ACTIVATION	400	The PSU has not activated the addressed account for the usage of the PIIS associated with the TPP.

3.20.5 Signing Basket Specific Error Codes

Message Code	HTTP Response Code	Description
REFERENCE_MIX_INVALID	400	The used combination of referenced objects is not supported in the ASPSPs signing basket function.
REFERENCE_STATUS_INVALID	409	At least one of the references is already fully authorised.

3.20.6 Push AIS specific Error Codes

Push Account Information service specific error codes are as follows:

Message Code	HTTP Response Code	Description
PRIOR_SUBSCRIPTION_AVAILABLE	409	The PSU/Corporate initiating the Push Account Information Service subscription already holds a subscription in the quadruple API Client/PSU/ASPSP/subservice or API Client/Corporate/ASPSP/subservice respectively or in the triple API Client/ASPSP/subservice in the case of direct access.
SUBSCRIPTION_INVALID	400	Return this code in case the API Client tries to add a subscription entry, delete a subscription entry or a subscription itself where it had been deactivated already before.
SECONDARY_URI_NOT_SUPPORTED	400	An API Client is trying to register a secondary Push URI within a Push Account Information subscription, which is not supported by the ASPSP.
MIME_TYPE_NOT_SUPPORTED	400	An API Client is trying to define a Push Account Information subservice with one or several mime types, where none of the mime types is

Message Code	HTTP Response Code	Description
		supported by the ASPSP for this subservice.
LIMIT_FOR_BALANCE_NOT_SUPPORTED	400	The subscription for a Push Balance Information Subservice is asking to involve a credit limit into the balance selection criteria, where the related balance does not support to include a related credit limit of the account.
ACCOUNT_CURRENCY_NOT_MATCHING	400	The API Client initiated a subscription where an amount is involved which is not matching the currency of the addressed account.
BALANCE_TYPE_NOT_SUPPORTED	400	The API Client initiated a subscription for a balance type that is not supported for the addressed account by the ASPSP.

This data type is used by [oFA-PFSM].

3.21 Mime Type Code

Code	Description
application/JSON	Mime type for body encoding in JSON syntax.
application/XML	Mime type for body encoding in XML syntax.
plain/text	Mime type for body encoding in text.

3.22 Name Prefix Code

Specifies the terms used to formally address a person.

Code	Description
DOCT	Doctor

Code	Description
MADM	Madam
MISS	Miss
MIST	Mister
MIKS	GenderNeutral

3.23 Onboarding Status

Code	Description
received	The first POST has been received and the onboarding resource has been created. The check of the input parameter was successful. The services identified by servicelds belong to the scheme identified by the schemelds. The API Client is known. The necessary service contract has been signed and correspond to the ID given by serviceContractId. The confirmation of the price lists is necessary and not done until now.
rejected	The second step (PUT) with the confirmation of the price lists has been failed. This can be e.g. due to one of the following reasons: • apiClient not known or no match with QWAC. • Necessary confirmation of conditions failed.
expired	The status of the onboarding resource is set to expired on day validTo+1.
terminatedByApiClient	The status of the onboarding resource is set to terminatedByApiClient, if a DELETE request has been sent by the API Client.
terminatedByAspsp	The status of the onboarding resource is set to terminatedByAspsp, if the ASPSP has terminated the corresponding onboarding.

Code	Description
valid	The onboarding of the API Client to the services identified by serviceIds has been completed successfully. No changes of the validity since the onboarding has been completed.
validTimeRestricted	The onboarding of the API Client to the services identified by serviceIds has been completed successfully. The validity has been changed since the onboarding has been completed. The validTo parameter has been set to a date when the onboarding will expire. Reasons for this will be defined by the business logic.

This data type is used in [oFA-IG-Adm].

3.24 Order Status Code

Code set to identify the status of an order:

Code	Description
unknown	The status of this order can currently not be determined.
new	Outstanding order with no executions.
partiallyFilled	Outstanding order with executions and remaining quantity.
filled	Order completely filled, no remaining quantity.
doneForDay	Order not, or partially, filled; no further executions forthcoming for the trading day.
canceled	Cancelled order with or without executions.
replaced	Cancelled order due to a replacement with or without executions.
pendingCancel	Order with a request for cancellation pending. Does not indicate that the order has been cancelled.
stopped	Order has been stopped at the exchange. Used when guaranteeing or protecting a price and quantity.
rejected	Order has been rejected by sell-side (broker, exchange, ECN).

Code	Description
	Note: An order can be rejected subsequent to order acknowledgment, i.e. an order can pass from "new" to "rejected" status.
suspended	Order has been placed in suspended state at the request of the client.
pendingNew	Order has been received by sell-side's (broker, exchange, ECN) system but not yet accepted for execution.
calculated	Order has been completed for the day (either filled or done for day). Commission or currency settlement details have been calculated.
expired	Order has been cancelled in broker's system due to time in force instructions.
acceptedForBidding	Order has been received and is being evaluated for pricing.
pendingReplace	Order with an Order Cancel/Replace Request pending.

3.25 Order Time Limit Code

Code set to identify the time an order shall be in force. Possible values:

Code	Description
day	Day Order: Limit order that is only good for the remainder of the trading day on which it is placed.
goodTillCancel	Good till cancelled (GTC) Order: Order that is working regardless of the time frame, until the order is explicitly cancelled. (Might still be subject of time limits from the ASPSP / broker).
atTheOpening	At the Opening Order: Order to be executed at the very beginning of the trading day.
immediateOrCancel	Immediate or cancel (IOC) Order: Order to be executed immediately and any unfilled portion be cancelled.
fillOrKill	Fill or kill (FOK) Order: Order that is directed to be executed immediately at the market or a specified price or canceled if not filled.

Code	Description
fillAndKill	Fill and kill Order: Order that can be filled partially or cancelled. If the order cannot be filled in its entirety at the specified premium, as much as possible of the order is filled and the rest is killed.
goodTillCrossing	Good till Crossing (GTX) Order: Order that is canceled prior to the market entering into an auction, or crossing phase.
goodTillDate	Good till Day or Date (GTD) Order: Order which remains active till its specified date, unless it has already been fulfilled or cancelled.
atTheClose	At the Close Order: Order to be executed at the end of the trading day, at the price available.
goodThroughCrossing	Good through Crossing Order: Order that is valid up till and including a crossing phase.
atCrossing	At Crossing Order: Order that is valid only during crossing (auction) phases.
goodForTime	Good for Time (GFT) Order: Order that is valid for a pre-defined time period.
goodForAuction	Good for Auction (GFA) Order: Order that is valid for an auction initiated by a trading firm.
goodForMonth	Good for this Month (GFM) Order: Order that is valid until the end of the current month, i.e. from the time of order submission until the end of the last trading day of the current month.

3.26 Payment Product

The definition of payment products as used in URIs for payment initiation. The following table might be extended on domestic level. The encoding of the codes is Max35Text.

Attribute	Description
sepa-credit-transfers	
instant-sepa-credit-transfers	

Attribute	Description
cross-border-credit-transfers.	
target-2-payments	

3.27 Preferred Method

Preferred method used to reach the contact.

Code	Description
LETT	Letter
MAIL	Email
PHON	Phone
FAXX	Fax
CELL	MobileOrCellPhone

3.28 Priority Code

This code set is equivalent to the ISO 20022 definition Priority2Code:

Type	Description
HIGH	Priority level is high.
NORM	Priority level is normal.

This code is used in [oFA-PFDom].

3.29 Referred Document Type Code

This code set is equivalent to the ISO 20022 definition DocumentType6Code:

Type	Description
MSIN	Document is an invoice claiming payment for the supply of metered services, for example gas or electricity supplied to a fixed meter
CNFA	Document is a credit note for the final amount settled for a commercial transaction.
DNFA	Document is a debit note for the final amount settled for a commercial transaction.
CINV	Document is an invoice.
CREN	Document is a credit note.
DEBN	Document is a debit note.
HIRI	Document is an invoice for the hiring of human resources or renting goods or equipment.
SBIN	Document is an invoice issued by the debtor.
CMCN	Document is an agreement between the parties, stipulating the terms and conditions of the delivery of goods or services.
SOAC	Document is a statement of the transactions posted to the debtor's account at the supplier.
DISP	Document is a dispatch advice.
BOLD	Document is a shipping notice.
VCHR	Document is an electronic payment document.
AROI	Document is a payment that applies to a specific source document.
TSUT	Document is a transaction identifier as assigned by the Trade Services Utility
PUOR	Document is a purchase order.

3.30 Regulatory Reporting Type Code

Type	Description
CRED	Regulatory information applies to the credit side.
DEBT	Regulatory information applies to the debit side.
BOTH	Regulatory information applies to both credit and debit sides.

3.31 Request Status

Syntactically, the same codes are used as in Transaction Status, but with a different semantic, since the codes are related to a Request To Pay transaction and **not** to the corresponding payment.

Code	Description
ACCP	The request to pay has been explicitly accepted by the PSU.
ACWC	The request to pay has been explicitly accepted by the PSU with changes related to the execution date or the accepted amount.
RCVD	The request to pay data can be mapped to a PSU, no restrictions of the PSU related to payment requests apply and the payment request can be further processed.
PDNG	- The request to pay transaction has not yet been mapped to a PSU, or - PSU related restrictions regarding request to pay functionality could not yet been checked
RJCT	The request to pay has been rejected by either the API Server, the ASPSP or the PSU.

This code is used in [oFA-IG-RTP].

3.32 SCA Approach Type

Code	Description
Decoupled	Decoupled SCA Approach
Embedded	Embedded SCA Approach

Code	Description
OAuth2	Redirect SCA Approach based on OAuth2 protocol
Redirect	Plain Redirect Approach

3.33 SCA Process Flow Requirement Code

Attribute	Description
psuAuthenticationRequired	The ASPSP will not select the SCA approach to be applied until the TPP has successfully updated the PSU Authentication or delivered an OAuth2 access token.
psuIdentificationRequired	The ASPSP mandates the PSU-ID for services involving SCA (like PIS, AIS, ...) before offering a SCA Selection.
scaMethodSelectionRequired	The ASPSP will not select the SCA approach to be applied until the PSU has selected the authentication method to be applied via the API Client. Note: As the PSU is neither authenticated nor identified at this point, the ASPSP will offer the ASPSP-specific SCA methods. Accordingly, the PSU has to choose a method which is allowed to him.
none	The ASPSP is able to select the SCA approach to be applied without a preceding PSU Authentication, PSU Identification or Authentication Method Selection. This then could also be handled in the online channel during authorisation.

3.34 SCA Exemption Code

Code	Description
none	no SCA exemption supported for payment initiation.
lowValueTransactions	
paymentsToSelf	
transactionRiskAnalysis	
trustedBeneficiaries	

3.35 SCA Status

The following codes are defined for this data type.

Remark for Future: A rework of the coding will follow, first Codes are given below:

Code	Description
received	An authorisation or cancellation-authorisation resource has been created successfully.
psuIdentified	The PSU related to the authorisation or cancellation-authorisation resource has been identified.
psuAuthenticated	The PSU related to the authorisation or cancellation-authorisation resource has been identified and authenticated e.g. by a password or by an access token.
scaMethodSelected	The PSU/TPP has selected the related SCA routine. If the SCA method is chosen implicitly since only one SCA method is available, then this is the first status to be reported instead of "received".
started	The addressed SCA routine has been started.
unconfirmed	SCA is technically successfully finalised by the PSU, but the authorisation resource needs a confirmation command by the TPP yet.
finalised	The SCA routine has been finalised successfully (including a potential confirmation command). This is a final status of the authorisation resource.
failed	The SCA routine failed. This is a final status of the authorisation resource.
exempted	SCA was exempted for the related transaction, the related authorisation is successful. This is a final status of the authorisation resource.

This data type is used in [oFA-PFSM].

3.36 Securing Method Code

This code is providing information about how an authorised amount is secured within the ASPSP.

Code	Description
reservationOfFunds	The bank is reserving the funds via a disposition of the amount on the account.
creditLine	The bank is reserving the funds via the usage of a credit line of the customer.
bankGuarantee	The bank is offering a payment guarantee.
consumerLoan	The payment is secured by a consumer loan.

This code is used in [oFA-IG-EPIS].

3.37 Securities Fee Type Code

This code is providing information the type of a fee that is applicable to a securities account.

Code	Description
transactionFee	Fee that applies to a transaction within this securities account.
brokerageFee	Fee a broker charges to execute transactions or provide specialized services on behalf of clients.
managementFee	Fee levied by an investment manager for overseeing an investment fund.
courtage	Courtage
custodyFee	Fee charged by the ASPSP for safekeeping the securities within the securities account.
exchangeFee	Fee charged for transferring shares from one fund to another fund within the same group.
thirdPartyFee	Fees charged by a third party
otherFee	Fee that is not covered by any of the fee types above.

3.38 Securities Order Side

Code set to identify the "side" / intended action for the respective order. Possible values:

Code	Description
buy	Order to buy.
sell	Order to sell.
subscription	Order for a subscription.
redemption	Order for a redemption.

3.39 Service Type

Code	Description
PIS	single payments
RPIS	periodic payments
BPIS	bulk payments
XFPIIS	secured-payments
XDPIS	deferred-payments
XDFPIS	secured-deferred-payments
XMDPIS	multiple-deferred-payments
XMDFPIS	secured-multiple-deferred-payments
XRPIIS	recurring-payments
XRFPIS	secured-recurring-payments
XMRPIS	multiple-recurring-payments
COF	funds confirmation
CONS-AIS	consent on account information
XCONS-DOC	consent for document services
XCONS-PSUP	consent on psu parameter information
XCONS-COF	consent for acceptance of COF requests

Code	Description
AIS	Account information for payment accounts
XAIS-LA	Account information for loan accounts
XAIS-SA	Account information for savings accounts
XAIS-CA	Account information for single cards
XAIS-SC	Account information for securities accounts
XPAIS-AE	Push account information service for account entries
XPAIS-AB	Push account information service for account balances
XPAIS-AS	Push account information service for account statements
XPAIS-RTP	Push account information service for incoming RTP
RTP	Request to Pay

This code is used in [oFA PAIS]

3.40 .Signing Class

The definition of authorisation rights of a user.

Attribute	Description
singleSigning	User is entitled to sign related transactions on his/her own.
firstSigning	User is entitled to sign related transactions as a first signer.
secondSigning	User is only entitled to sign related transactions after a first signer has signed the transaction.
nonSpecifiedSigning	User is entitled to sign related transactions, without providing more detailed information about the related signing class.
transportSigning	User is only entitled to sign the submission of a related transaction, but not to authorise the related transaction.
none	User has no rights related submission or authorisation of related transactions.

3.41 Status Reason Code

At a minimum, the following codes from the ISO20022 ExternalStatusReasonCode1 list need to be supported by the ASPSP. The codes should only be used, if a related resource has been generated on API level, i.e. if a transaction status can be provided. Otherwise a related Message Code to be used in Error information would be sufficient.

Code	Name	ISO 20022 Definition and openAPI Remarks
AM04	InsufficientFunds	Amount of funds available to cover specified message amount is insufficient. Remark: Indicates that the reason for rejecting the payment is that the required funds have been found to be not available for the specific (e.g. due to missing funds or due to configured limits) during processing after the initial acceptance of the payment initiation.
DS0C	SignerCertificateRevoked	The signer certificate is revoked. Remark: Is used, if the signature/corporate seal certificate is revoked.
DS0D	SignerCertificateNotValid	The signer certificate is not valid (revoked or not active). Remark: Is used, if the signature/corporate seal certificate is blocked by the ASPSP or if the certificate is expired.
DS0A	DataSignRequested	Data signature is required. Remark: Is used, if a signing by a corporate seal/certificate is mandated by the ASPSP but is missing.
DS04	OrderRejected	The order was rejected by the bank side (for reasons concerning content) Remark: Indicates that the reason for rejecting the payment is that the content of the payment initiation has been found invalid during processing after the initial acceptance of the payment initiation.
SL11	CreditorNotOnWhitelistOfDebtor	Whitelisting service offered by the Debtor Agent; Debtor has not included the Creditor on its "Whitelist" (yet). In the Whitelist the Debtor may list all allowed Creditors to debit Debtor bank account.

This code is used in [oFA-IG-Com], [oFA-PFSM]..

3.42 Subscription Status

Code	Description
received	The subscription data have been received and are technically correct. The data is not authorised yet.

Code	Description
rejected	The subscription data have been rejected e.g. since no successful authorisation has taken place.
partiallyAuthorised	The subscription is due to a multi-level authorisation, some but not all mandated authorisations have been performed yet.
valid	The subscription is accepted and valid for the related subscribed services as specified in the subscription object.
validInChange	The subscription is accepted and valid for the related subscribed services as specified in the subscription object. A new subscription entry has been submitted which is not yet (fully) authorised.
revokedByPsu	The subscription has been revoked by the PSU towards the ASPSP.
cancelledByAspsp	The subscription has been cancelled by the ASPSP.
expired	The subscription expired.
terminatedByTpp	The corresponding TPP has terminated the subscription by applying the DELETE method to the subscription resource.

The ASPSP might add further codes. These codes then shall be contained in the ASPSP's documentation of the XS2A interface.

This code is used in [oFA-PFSM], [oFA PAIS].

3.43 Subscription Entry Status

Code	Description
received	The subscription data have been received and are technically correct. The data is not authorised yet.
rejected	The subscription data have been rejected e.g. since no successful authorisation has taken place.
partiallyAuthorised	The subscription is due to a multi-level authorisation, some but not all mandated authorisations have been performed yet.
valid	The subscription is accepted and valid for the related subscribed service as specified in the subscription object.

Code	Description
revokedByPsu	The subscription has been revoked by the PSU towards the ASPSP.
expired	The subscription expired.
terminatedByTpp	The corresponding TPP has terminated the subscription by applying the DELETE method to the subscription resource.

This code is used in [oFA-PFSM], [oFA PAIS].

3.44 Trading Session Type Code

Code set to identify the type of a trading session, in which an order shall be executed. Possible values:

Code	Description
auctions	Orders that are traded in regularly occurring auctions: orders are accumulated and executed on the basis of defined algorithms.
continuous	Orders that are executed directly against each other as they hit the marketplace.

3.45 Transaction Status

The transaction status is filled with codes of the ISO 20022 data table:

Code	Name	ISO 20022 Definition
ACCC	AcceptedSettlementCompleted	Settlement on the creditor's account has been completed.
ACCP	AcceptedCustomerProfile	Preceding check of technical validation was successful. Customer profile check was also successful.

Code	Name	ISO 20022 Definition
ACSC	AcceptedSettlementCompleted	Settlement on the debtor's account has been completed. Usage: this can be used by the first agent to report to the debtor that the transaction has been completed. Warning: this status is provided for transaction status reasons, not for financial information. It can only be used after bilateral agreement
ACSP	AcceptedSettlementInProgress	All preceding checks such as technical validation and customer profile were successful and therefore the payment initiation has been accepted for execution.
ACTC	AcceptedTechnicalValidation	Authentication and syntactical and semantical validation are successful
ACWC	AcceptedWithChange	Instruction is accepted but a change will be made, such as date or remittance not sent.
ACWP	AcceptedWithoutPosting	Payment instruction included in the credit transfer is accepted without being posted to the creditor customer's account.
RCVD	Received	Payment initiation has been received by the receiving agent.
PDNG	Pending	Payment initiation or individual transaction included in the payment initiation is pending. Further checks and status update will be performed.
RJCT	Rejected	Payment initiation or individual transaction included in the payment initiation has been rejected.
CANC	Cancelled	Payment initiation has been cancelled before execution
ACFC	AcceptedFundsChecked	Pre-ceeding check of technical validation and customer profile was successful and an automatic funds check was positive .
PATC	PartiallyAcceptedTechnical Correct	The payment initiation needs multiple authentications, where some but not yet all have been performed. Syntactical and semantical validations are successful.
PART	PartiallyAccepted	A number of transactions have been accepted, whereas another number of transactions have not yet achieved 'accepted' status. Remark: This code may be used only in case of bulk payments. It is only used in a situation where all mandated authorisations have been applied, but some payments have been rejected.

This code is used in [oFA-IG-Com], [oFA-PFSM], [oFA-IG-RTP].

3.46 Transaction Type

Attribute	Description
payments	All payment types

Attribute	Description
payments.sepa-credit-transfers	All sepa credit transfers.
payments.cross-border-credit-transfers	Cross border payments.
consents	Consents, e.g. on AIS
subscriptions	subscriptions for Push AIS Services
all	All transaction types that need authorisation.

3.47 Type of Order Code

Code set to indicate characteristics of an order.

Code	Description
allOrNone	A round-lot market or limit-price order which must be executed in its entirety or not at all; unlike 'fill or kill', these orders are not cancelled if not executed as soon as received.
buyContraShortExempt	Order to buy contra short exempt.
buyContraShort	Order to buy contra short.
buyMinus	Order to buy at a price lower than the current market price. This is an order to buy a stated amount of a financial instrument provided that its price is not higher than the last sale if the last sale was a minus or zero minus tick, not higher than the last sale minus the minimum fractional change in the stock if the last sale was a plus or zero plus tick. The price limit indicates the highest price at which the order can be executed.
carefully	Order that is not to be executed as a whole because it may disturb the price.
combinationOrder	Order that is linked to another order to buy or sell and must be executed as a unit, both or none, or cancelled as a unit.
discretionary	Order where the executing broker or investment manager decides on the quantity or price.

Code	Description
doNotIncrease	Limit order to buy or stop order to sell or stop limit order that is not to be increased in shares on the ex-dividend date as a result of a stock dividend or distribution.
doNotReduce	Limit order to buy or stop order to sell, or stop-limit order to sell that is not to be reduced in price by the amount of an ordinary cash dividend. Applies only to ordinary stock dividends; should be reduced for other distributions.
icebergOrder	Type of limit order whose overall quantity is not transparent to the market. Rather, only a client-defined part of the order is shown to the market (the 'tip of the iceberg'). Upon execution of the first 'tip' (that is partial execution to the client while for the market it does not appear to be a 'partial') the system releases the next 'tip' of the same size to the market until the 'iceberg' has been melted down successfully. Different from a usual care order.
limitWith	Order to be executed at a limit price, with a round-lot (or board-lot) sales; valid only for odd lot orders.
limitWithout	Order to be executed at a limit price, without a round-lot (or board-lot) sales; valid only for odd lot orders.
limitOrder	Order to buy at the indicated price limit or lower or an order to sell at the indicated limit price or higher.
atMarket	Order to buy or sell a specified amount of a financial instrument at the quoted market price or better.
marketNotHeld	Order to buy or sell a specified amount of a financial instrument at the quoted market price or better with some discretion on the price limit.
marketToLimitOrder	Type of order that couples the high possibility of execution (Market Order) with a protection against unwanted price fluctuations (Limit Order).
marketUntilTouched	Order to buy or sell a specified amount of a financial instrument at the quoted market price or better with some discretion on the price limit.
notHeld	Order that may be executed in partials or outside the hours of the exchange or other exchange rules.

Code	Description
orderLie	Order that is related to another order where the second order may be cancelled without cancelling the first. Normally, the sell order must be executed before the buy order.
stopLimit	Stop order to buy (sell) that becomes a limit order at the limit price when the financial instrument trades at or above (below) the stop price after the order is submitted.
stopOrder	Order to buy that becomes a market order when the financial instrument trades at or above the stop price after the order is submitted or an order to sell which becomes a market order when the financial instrument trades at or below the stop price.
stopLoss	Order to sell that sets the sell price below the market price.
sellPlus	Order to sell a stated amount provided that the price is not lower than the last sale price if the last sale was a plus or zero plus tick and not lower than the last sale minus the minimum fractional change in the financial instrument if the last sale was a minus or zero minus tick.
sellShortExempt	Order to sell short which is exempt from short-sale rules.
sellShort	Order to sell a financial instrument that the seller does not own; a sale effected by delivering a financial instrument borrowed by or for the account of the seller.

3.48 Other ISO-related basic Types

The following codes and definitions are used from ISO 20022

- **Account Identification Code:** ExternalAccountIdentification1Code
- **Address Type Code:** AddressType2Code
- **Bank Transaction Code:** ExternalBankTransactionDomain1Code
- **BBAN:** BBANIdentifier
- **BICFI:** BICFIIdentifier
- **Category Purpose Code:** ExternalCategoryPurpose1Code
- **Cash Account Type:** ExternalCashAccountType1Code

- **Clearing System Identification Code:** ExternalClearingSystemIdentification1Code
- **Document Format Code:** ExternalDocumentFormat1Code
- **Document Type Code:** ExternalDocumentType1Code
- **Exchange Rate Type Code:** ExchangeRateType1Code
- **External Financial Instrument Identification Type Code:** ExternalFinancialInstrumentIdentificationType1Code (see [ISO20022_SR])
- **External Local Instrument Code:** ExternalLocalInstrument1Code
- **Financial Institution Code:** ExternalFinancialInstitution1Code
- **IBAN:** IBAN2007Identifier
Pattern: [A-Z]{2}[0-9]{2}[A-Z0-9]{1,30}
- **Instruction Code:** Instruction3Code
- **ISIN:** ISINOct2015Identifier (see [ISO20022_SR])
Pattern: [A-Z]{2,2}[A-Z0-9]{9,9}[0-9]{1,1}
- **Language Code:** Code following ISO – 639-3 (three letter code)
- **LEI:** LEIIdentifier as defined in ISO 17442 "Financial Services - Legal Entity Identifier (LEI)".
- **Mandate Reason Code:** ExternalMandateReason1Code
- **Market Type Code:** MarketType4Code (see [ISO20022_SR])
- **Merchant Category Code:** Category code conform to ISO 18245
- **Organisation Identification Code:** ExternalOrganisationIdentification1Code
- **PersonIdentificationCode:** ExternalPersonIdentification1Code
- **Phone Number:** PhoneNumber
- **Purpose Code:** ExternalPurpose1Code
- **Proxy Account Type Code:** ExternalProxyAccountType1Code
- **Sequence Type Code:** SequenceType3Code

- **Service Level Code:** ExternalServiceLevel1Code
- **Transaction Activity Code:** TransactionActivity1Code (see [ISO20022_SR])
- **Type of Price Code:** TypeOfPrice17Code (see [ISO20022_SR])

The following code is a concatenated code from ISO20022

- **BankTransactionCode:** This code type is concatenating the three ISO20022 Codes Domain Code, Family Code and SubFamily Code by hyphens, resulting in "DomainCode"- "FamilyCode"- "SubFamilyCode".

Example: PMNT-RCDT-ESCT defining a transaction assigned to the PayMeNT Domain (PMNT), belonging to the family of ReceivedCreDitTransfer (RCDT) that facilitated the EuropeanSEPACreditTransfer (ESCT)

To reflect a securities position's balance type, **SecuritiesBalanceTypeCode** is defined as the code list, containing all codes from ISO 20022 code lists

- SecuritiesBalanceType1Code (see [ISO20022_IF])
- SecuritiesBalanceType2Code (see [ISO20022_IF])
- SecuritiesBalanceType11Code (see [ISO20022_SR])
- SecuritiesBalanceType12Code (see [ISO20022_SR])

For all codes used in JSON structures, not the abbreviation defined for XML encoding, but the name of the code is used as value.

The following Codes are used from other ISO standards:

- **Currency Code:** Codes following ISO 4217 Alpha 3
- **Country Code:** Two characters as defined by ISO 3166

Further basic ISO data types:

- **ISODatetime:** A particular point in the progression of time defined by a mandatory date and a mandatory time component, expressed in either UTC time format (YYYY-MM-DDThh:mm:ss.sssZ) or local time with UTC offset format (YYYY-MM-DDThh:mm:ss.sss+/-hh:mm),. These representations are defined in "XML Schema Part 2: Datatypes Second Edition - W3C Recommendation 28 October 2004" which is aligned with ISO 8601.
- **ISODate:** A particular point in the progression of time in a calendar year expressed in the YYYY-MM-DD format.

3.49 Additional Cryptographic Code Lists

Codes to identify algorithms for signature generation and encryption shall be used as defined by [RFC7518]. At least the following codes have to be supported:

Signature Algorithm Code

PS256

for RSASSA-PSS using SHA-256 and MGF1 with SHA-256 (section 3.5 of [RFC7518])

Key Encryption Algorithm Code

RSA-OAEP-256

for RSAES OAEP using SHA-256 and MGF1 with SHA-256 (section 4.3 of [RFC7518])

Content Encryption Algorithm Code

A256GCM

for AEC in Galois/Counter Mode using 256-bit key (section 5.3 of [RFC7518])

4 References

4.1 Normative References

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- [eIDAS] Regulation (EU) No 910/2014 of the European Parliament and of the Council on Electronic Identification and Trust Services for Electronic Transactions in the Internal Market, 23 July 2014, published 28 August 2014
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- [RFC5843] Bryan, A, "Additional Hash Algorithms for HTTP Instance Digests", RFC 5843, DOI 10.17487/RFC5843, April 2010, <https://www.rfc-editor.org/info/rfc5843>
- [RFC6749] Hardt, D., "The OAuth 2.0 Authorization Framework", October 2012, <https://tools.ietf.org/html/rfc6749>
- [\[RFC7231\] R. Fielding, J. Reschke, Hypertext Transfer Protocol \(HTTP/1.1\): Semantics and Content](#)
- [RFC7518] JSON Web Algorithms (JWA), Request for Comments 7518, Internet Engineering Task Force (IETF), M. Jones (Microsoft), May 2015
- [RFC7807] M. Nottingham, Akamai, E. Wilde, „Problem Details for HTTP APIs“, March 2016, <https://tools.ietf.org/html/rfc7807>
- [\[RFC 8414\] M. Jones, N. Sakimura, J. Bradley; "OAuth 2.0 Authorization Server Metadata";](#)

4.2 Informative References

- [oFA-IG-Com] openFinance API Framework, Implementation Guidelines, Compliance Services, Version 2.0, 05 October 2023.
- [oFA-IG-EPIS] openFinance API Framework, Implementation Guidelines, Extended Payment Initiation Services, Version 2.1, 31 July 2024.
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- [oFA-IG-RTP] openFinance API Framework, Implementation Guidelines, Request to Pay Services, version 2.0, November 2023
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5 Annex A: Change Log

5.1 Changes from Version 2.1 to Version 2.2

The following changes have been applied in version 2.2 relative to version 2.1:

Section	Change	Reason
Preamble	The licence conditions of Berlin Group have been removed.	To support the European community in re-using the openFinance data model, when not directly using openFinance APIs.
2.1	The attribute cardAccounts have been added.	single card accounts and reconciliation card accounts need now to be addressed explicitly in the consent. The former solution did not allow a proper differentiation. Erratum
2.7	The attribute resourceId has been restricted from string to max70Text.	Clarification, as applied everywhere in the API Framework.
2.7	The description of the attribute ownerNames was extended.	Clarification
2.16	New data type to support the features of adding a debtor account to a resource.	CR 113
2.43	The spelling of the attribute clearingSystemIdentificationCode was corrected.	Erratum
2.67	Added the status reason to the initiation data attributes	Clarification.
2.71	Added an attribute to "Links" for indicating an update of a resource by a debtor account.	CR113
2.78	The attributes oauth2PrestepMetadataUri and oauth2OfflineScopeRequired have	Better support of documentation.

Section	Change	Reason
	been added to the API Security parameters.	
2.80	Added the value "pageSize" for supported query parameters in transactions	CR115
2.80	Added the attribute maxPageSize	CR115
2.103	Some attributes adapted to lowerCamelCase	Errata
3.48	Do not allow pure local time zone anymore (only with UTC offset).	CR111

5.2 Changes from Version 2.2 to Version 2.2.1

The following changes have been applied in version 2.2.1 relative to version 2.2:

Section	Change	Reason
2.50	New attribute "Duration" for Mandate API	Support of new service
2.56	New attribute "Extended Party Description" for Mandate API	Support of new service
2.73	New attribute "Mandate Type" for Mandate API	Support of new service
2.74	New attribute "Mandate Status" for Mandate API	Support of new service
2.76	New attribute "Occurrences" for Mandate API	Support of new service
2.77	Condition of attributes changed	Errata
2.122	New attribute "Referred Mandate Document information"	Support of new service

Section	Change	Reason
3.8	New "Classification code" for Mandate API	Support of new service
3.19	New "Mandate Setup Reason Code" for Mandate API	Support of new service
3.20.1	New codes "CLIENT_INVALID" and "CLIENT_INCONSISTENT"	Support error information for directory related issues
3.48	Added "Organisation Identification Code" and "Mandate Reason Code"	Errata and support of new service